

Ministero dell'Università e della Ricerca
Direzione Generale della Ricerca

FONDO ITALIANO PER LE SCIENZE APPLICATE (FISA)
AVVISO PER LA PRESENTAZIONE DI PROPOSTE PROGETTUALI

Plan4ARI

Planning for Artificial Robotic Intelligence and Beyond

Project details

Acronym	Plan4ARI
Title	Planning for Artificial Robotic Intelligence and Beyond
Duration (months):	48
Keywords (max. 5 words)	Advanced Robotics; Motion Control; Task Planning; Low-Code; Natural Interaction
Project Area	Advanced Manufacturing
Principal Investigator (PI)	Nicola Pedrocchi
Host Institution (HI)	COMAU SpA
Affiliation Institution (AI)	Consiglio Nazionale delle Ricerche
Research Organization	Consiglio Nazionale delle Ricerche

Abstract

We aim to ensure that artificial robotic intelligence (**ARI**) —AI-powered robotic systems — benefits all advanced manufacturing areas. If **ARI** is successfully created, this technology could dramatically improve their shopfloor productivity and aid in adopting industrial robots in application areas far from the actual market. **ARI** has the potential to give robots incredible new capabilities; we can imagine a shopfloor where all the robots have access to help with almost any cognitive task, providing a significant force multiplier for human creativity and easing human working conditions. The project will approach all the building bricks over which a robot controller is designed. We will investigate and innovate on adopting AI in high-rate real-time controllers, motion planning, task planning, and user-robot communication. On the other hand, **ARI** would also come with a severe risk of misuse, drastic accidents, and societal disruption. The project will deeply investigate the dark side of the deep usage of AI in industrial robotic systems. The generative AI tools will be exploited for the user-robot interfaces to ease the programming and robot adoption. Innovative deliberative AI techniques will be integrated with the high-level interfaces to generate safe task allocation and scheduling for any application. Deep learning methodologies will be integrated with sample-based innovative motion planners to update the target trajectory according to the scene continuously. 3D vision and 3D radar sensors will be used to keep the scenes updated, especially in collaborative applications with the robots. Fast neural controllers based on advanced AI methodologies will be envisaged to deploy fast and effective new generations of industrial controllers to improve robot performance dramatically.

A Project description

A.1 From Artificial Generic Intelligence (AGI) to Artificial Robotic Intelligence (ARI) for the Industry5.0 revolution

Artificial General Intelligence (AGI) is an artificial intelligence that can understand, learn, and apply knowledge across a wide range of tasks at a human-like level. Unlike narrow AI, designed to perform specific tasks, AGI would have the ability to think, reason, and solve problems in a generalized way, like human cognitive abilities. Several key features characterize AGI: **Versatility, Learning and Adaptation, Reasoning and Problem-Solving, Common Sense and Understanding, and Self-improvement**. Significant challenges, such as understanding human cognition, creating adaptable learning algorithms, and developing the computational power required, must still be overcome despite the high quality reached recently by tools like ChatGPT4 o1 and Claude 3.5 Sonnet and Gemini.

AGI is commonly considered the edge technology that will eventually achieve the ambition of **cognitive robotics**, which has been the ultimate research goal for the last two decades in industrial robotics and plays a fundamental role in the success of Industry5.0. The recent revolution of the **LLMs** is opening the question of **when** robots can integrate such tools and become **cognitive agents** in manufacturing.

As a human being, however, the robot is not only **AI fleshed out**, but the cognitive abilities rely on conceiving a multiple interacting system composed of different tools and agents. **Plan4ARI** aims to step towards a vast scientific leap, following a **systematic approach**.

Plan4ARI identifies four levels where AI can dramatically improve the usage of robots in an industrial context.

- A robot is intelligent if the user can describe the task to be tackled instead of coding
- A robot is intelligent if it can autonomously solve how to tackle a task, considering other agents (robots or humans) inside the operative scene
- A robot is intelligent if it can plan its movement considering the context and preserving safety under all the operative conditions
- A robot is intelligent if it can move and react appropriately during the execution of tasks according to the scene evolution

As humans solve the different steps using different capabilities of the neuronal system, **Plan4ARI** will conceive and deploy a new generation of intelligent robots integrating the most advanced cognitive solutions at the state of the art and beyond. **Artificial**

Robotic Intelligence (ARI) is a complex system where cognitive modules interact and cooperate to deploy the **best robotic companion** for human workers.

A.2 The Plan4ARI architecture and the COMAU controller

A.2.1 Context

Imagine a mobile manipulator with a high payload that must perform several welding applications in a fully unstructured environment like a huge boat. The working environment is very harmful and unhealthy for human workers. The references are weak, and many operations must be defined onsite.

Imagine a human operator controlling several different mobile robots inside this context. The operator should be able to describe the task to the robot by using natural language. Then, they should be enabled to see the virtual execution of the task. They should be able to notify of errors and eventually ask for plan refinement. Finally, they should execute the task and be notified if something unexpected happens. They should be able to move across the robots safely to check ongoing progress. **This is not an image of the future** since most fundamental branches are available at the state of the art and technology. They must only be refined and turned inside a new generation of robotics setup enabled with cognitive capabilities. **Natural language to task planning is at the edge of the revolution of Generative AI tools.** Technically, multi-agent generative AI with Reinforcement Learning (such as ChatGPT o1 or Claude 3.5 AI Sonnet) turns the request into a sequence of actions, solving the problem when described. These tools need proper prompt engineering to improve the quality of the result, and further tools must be able to verify, check, and correct all the errors that generative techniques introduce. Adopting deliberative AI tools to lint the output of generative AI techniques is the frontier of the research to achieve **safe, efficient, and optimal task planning.**

AI dramatically improves fully autonomous mobile manipulators' motion plan computation and scene registration. The AI-powered motion planner can transform symbolic goals into geometric targets, extrapolating the information from the images - with minimal support from the human user - and exploiting advanced localization and registration tools to make the robot aware of where it is inside the scene. The humans in the scene should be identified and tracked with compliance with all the safety regulations, and their intention should be estimated to optimize the motion plan and, consequently, the task plan.

AI-powered robots are expected to achieve a step-change in performance while continuously modifying the trajectory according to the scene evolution and the runtime quality tracker. The controller must integrate fast full-state observers to boost the control loop and minimize different functional goals according to the property of the task. The online modification of the velocity and the compensation of external dynamic load are still on the edge of technology. Integrating learning methodologies will allow for keeping a minimum dimensional dynamic model to control all the elastic and unmodelled dynamics.

A.2.2 The conceptual architecture

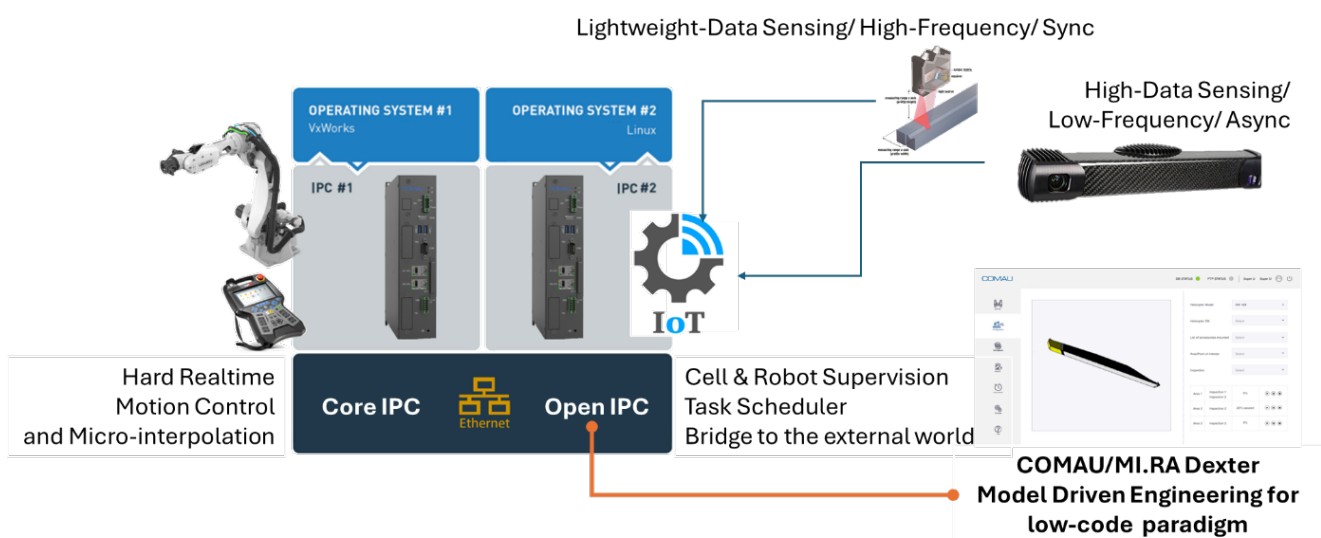


Figure 1 The Architecture for the new generation of robot controllers needs 2 IPC – The OPEN IPC provides all the programming tools, the motion planner, and the bridge to the world (cell & sensing). The Core IPC oversees the control loops and the micro-interpolation (high-rate real-time motion planner) and implements all the safety rules.

Implementing **Artificial Robotic Intelligence** requires an architecture with two control units, as in Figure 1.

First, the **Open IPC** will be devised with a cognitive programming platform, the **MI.RA/Dexter**¹ from COMAU. This platform is deeply under development in COMAU and allows two main options for the user: first, it can be fully customized; second, it

¹COMAU has designed MI.RA/Dexter in the last three years to be a platform capable of simplifying the programming of a production line, allowing the complete management of a robotic machine from software through the creation of a workflow. Dexter can program a line from scratch, integrate

consists of a block diagram programming interface (**model-driven engineering paradigm**). This block diagram programming allows the system to store a repository of completed algorithms, which can be put in a cascade to create a customized process pipeline. Those available algorithms represent atomic software functionalities (from simple tasks, such as changing I/O values, to complex tasks, like collision-free and optimized path planning) that inherit and extend abstract classes and methods to be compatible with the underlying software platform. This, in turn, allows the **process expert to be the system programmer since it can completely ignore activities such as user interface, camera calibration, etcetera**. The Model-Driven, therefore, opens a different cognitive interaction between human operators and systems, and it will allow in the future also for the integration of AI modules for the autonomous generation of process flows (exploiting the co-pilot functionalities offered by AI that are now in the hype for the code programmers). Using the classic **deliberative AI** nomenclature, the **MI.RA/Dexter** will be the **dispatcher** and the **execution monitor** engine. The PLC and all the automation will be connected to the robot controller and **MI.RA/Dexter** will deploy full virtualization for the User, controlling and monitoring the plan execution. **The Open IPC will bridge the robot's control and the external world**. It will implement the "firewall" between the non-real-time and real-time automation worlds. It will run powerful tools like ROS2, enabling the extension of the robot's capabilities by just connecting it to external nodes. **The Open IPC will implement the sensing and the motion planning interface**. It will provide powerful tools to plan the motion in every working condition autonomously.

Finally, the **motion controller** will run on the **Core IPC** to grant all the real-time performance and allow the implementation of advanced control architecture. Furthermore, the **Core IPC** will run an advanced hard-real time, high-rate motion planner that will allow the trajectory modification according to every working condition. The adoption of innovative strategies will revolutionize the lookahead.

Given such powerful architecture, **Plan4ARI** will, specifically:

- Integrate **Human-Robot Natural Communication** in the **MI.RA/Dexter** platform. It will consist of a module translating the **natural language** into a **verified and optimal flowchart**. **A powerful tool to reduce the coding to the minimum (e.g., zero for almost all situations)**
- **Plan4ARI** will expand **MI.RA/Dexter** with further modules for **scene/object registration/segmentation**. The **operative scene** will be **continuously updated**, and innovative fusion techniques to integrate new sensors (like radar) will be implemented
- **Plan4ARI** will expand **MI.RA/Dexter** with an innovative **motion planner** that will run **offline** and in **runtime**. The **points** will be continuously streamed from the **Core IPC** to the **Open IPC**, which will translate the flow of motion references into real-time setpoints for the **hard real-time motion controller**
- **Plan4ARI** will develop an innovative **Model Predictive Controller** for micro-interpolating incoming external references. This will revolutionize the lookahead approach in industrial robotics, allowing us to achieve impressive tracking performance at Cartesian and Joint levels. This innovation will connect the high-rate sensor directly to the motion controller to continuously grant the tracking performance in every working condition.
- **Plan4ARI** will develop an innovative **neural controller** for **motion control** with a **prescribed performance constraint** design. This will open a new generation of controllers that learn and tune the model, allowing the customization of the robotic controller on the system and impressively decreasing the deployment time.
- **Plan4ARI** will be natively developed in compliance with the **EU AI Act**, ensuring that all proposed AI solutions adhere to the highest standards of safety, transparency, and data protection required. The architecture will include specific measures to ensure critical AI systems are safe and reliable and respect fundamental rights.

A.2.3 Project's Modules

The Project organization is shown in the Figure below. **Plan4ARI** foresees four modules; each focused on solving a functional aspect of the cognitive architecture.

- **Low-code programming and Human-Robot Natural Communication**. This module will investigate all the aspects relative to how to program the robot using natural language, with the guarantee that the autonomously generated program is the best possible. This will be discussed in Section **A.3**.
- **Intelligent Sensing and Safe Motion Planning for Mobile Manipulators and Multi-Agents Interaction**. This module will innovate both in sensing the scene where the robot is operating and in motion planning. The two aspects will hardly be intertwined. The sensing will aim to generate a tool that allows a super/accurate registration of the robot in the scene and the semantic segmentation of the scene. The planner will tackle all the techniques related to path generation, considering all the scenarios: offline path generation, online re/planning, and high-rate safe hard-real time planning. This will be discussed in Section **A.4**.
- **Advanced Robot Motion Controlling through Machine Learning Methods**. This module will investigate a new generation of neural controllers to speed up the tuning and increase the tracking capability in all possible working conditions. This will

camera inputs into the system, process images through vision algorithms, move the robot via ROS or TCP/IP, manage collision avoidance, and more. The idea is to free the production process from the need for a specialized worker; the software's end user should be able, without having undergone a lengthy learning process, to perform the mentioned operations. **MI.RA/Dexter** aims to achieve a user-friendly platform.

be discussed in Section A.5.

- **Trustworthiness and Human-Centric Optimization.** This module will introduce tools to evaluate users' acceptance of AI-powered robots. AI could be scary if not correctly managed. The module will also investigate how to increase acceptance. This will be discussed in Section A.6.

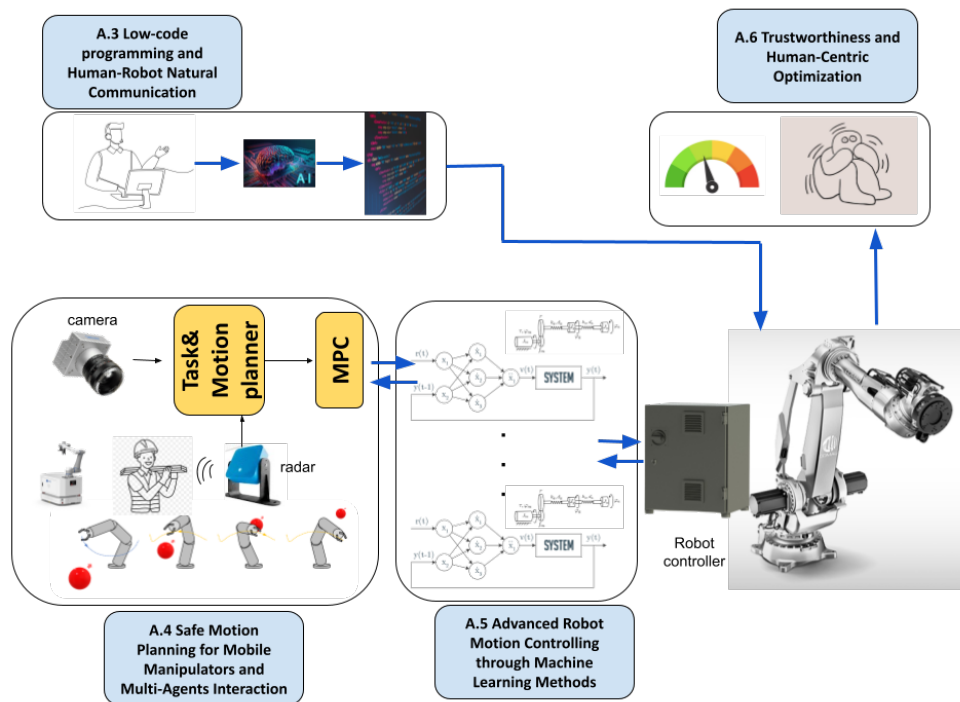


Figure 2 Graphical Representation of the Plan4ARI modules.

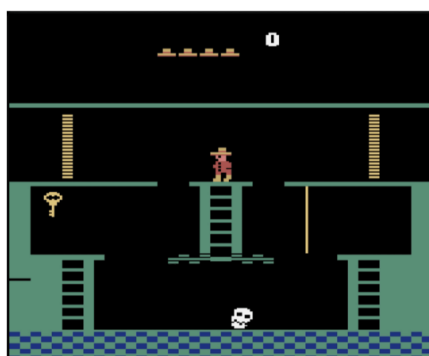
A.3 Low-code programming and Human-Robot Natural Communication

A.3.1 Objectives

The low-code approach usually asks the user to compose the part program by dragging and connecting functional modules that perform the sequence of actions needed to address a task. This approach is the core of the MI.RA/Dexter low-code programming tool from COMAU. The process expert usually considers the creation of such a program overwhelming. Model Driven Engineering is the basis of the low-code approach, and the user must know all about the libraries the interface has available. Furthermore, model-driven engineering requires that the user be skilled in sequential task allocation. Finally, the program composed by the User is often far from optimal, especially when multi-mobile manipulators are in the scene or when the human operators cooperate with the robots.

The new generation of cognitive robots should offer the possibility of **being programmed using unstructured natural language**. The user could be unaware of the robot's functionalities, automation possibilities, and PLC configurations, and they should be able to chat with the robot and check the robot's output using virtualization of the operations. Edge GenAI, like ChatGPT o1 or Claude AI 3.5 Sonnet, performs simple reasoning, as shown in the Figure below.

Language as a substrate for reasoning in neural networks



A man is standing on a platform suspended in the middle of the room. A ladder connects the platform he is standing on to another platform below. Next to this on the right, a rope is hanging off the ceiling. At the very bottom, on the ground floor, there is a skull rolling on the floor. On the left, on the ground floor, there is another ladder, and on top of that, there is a key. Describe the exact sequence of actions the man should perform.

- The man should reach the ladder below him and climb down to the bottom.
- The man should walk over to the rope on the right and climb up to the platform on the ceiling.
- The man should reach for the top of the ladder on the left and climb up to the platform on the ceiling.
- The man should reach for the key on the left and walk over to the right.
- The man should climb back down the ladder on the left.

Figure 3 Example of task planning and prompt engineering (credits to Giambattista Parascandolo, "Language as a substrate for reasoning", seminar at MIT, 2020, link: <https://sites.google.com/view/giambattista-parascandolo/home>).

Significant drawbacks are, however, still blocking the adoption of such technologies to task planning:

- The prompt must be appropriately engineered
- The output could be erroneous
- The plan is far from being optimum if the problem solution is not unique
- The plan cannot consider complex and variable operations

Plan4ARI aims to be disruptive in translating the natural language of the human operator into **Optimal Task Plans** ready to be executed. **Plan4ARI** will advance the SOTA by putting human operators at the center of the holistic control framework. Specifically, we will work to

- **Innovate methodologies for low-code programming.** The User should only be an expert in the process, not an expert in robot programming, and not an expert in the functions available in the SW. The user interface should enable communication of the desiderata and check errors using natural language.
- **Include the uncertainty of the task duration and the interaction,** exploiting data-driven metrics that allow for continuous interaction monitoring during the task execution and the plan's evolution to optimize objective cost function (like the plan execution).
- **Optimize task planning, especially for multi-robot systems, mobile manipulator teams, and AI-powered robotic companions.** The autonomously generated part program, which is ready to be executed, will also be optimized. It will consider minimizing the entire execution time or other heuristics.

A.3.2 Scientific and Technical Approach

A.3.2.1 Context and State-of-the-art

Turning natural language into a task plan is a fundamental research topic that has been heavily investigated recently. For brevity, a couple of paradigmatic works on literature are discussed to give context to the proposed approach.

Focusing on the translation of the natural language in a task plan, an important recent work² explores using LLMs by creating programming language-style prompts, including function definitions and environment descriptions. The system enhances PDDL generation by transforming high-level natural language instructions into actionable plans. The prompts are designed to fit the robot's capabilities and the current environment, ensuring compatibility between generated actions and the robot's capabilities. A further interesting work³ has extended such an approach to robotics applications. Specifically, the research aimed at transforming natural language instructions into task and motion planning (TAMP) for robots using models that combine symbolic planning with neural networks, like PDDL-based methods. It leverages deep learning to bridge language and robotic action sequences.

Such approaches, however, are not systematic and have many limitations in industrial applications, where robustness is fundamental. Indeed, they do not have deterministic behavior, and it is not possible to have hard-constrained solutions.

A different approach has been followed in "PDDLStream: Integrating Symbolic Planners and Blackbox Samplers via Optimistic Adaptive Planning."⁴ (2020) – that focuses on integrating symbolic planners, like those using PDDL, with LLM-based black box samplers for task and motion planning (TAMP). The goal is to generate plans using a combination of high-level language instructions and learned models for complex robotic tasks. Despite the high quality of PDDLStream, they weakly support temporal planning, i.e., the scenario where each task has a predefined duration, and they cannot address uncertainty in the durations.

A further inspiring work is "Constructing Behavior Trees from Temporal Plans for Robotic Applications."⁵ mainly focused on a framework for executing temporal plans in the real and open world, which requires adapting to environmental uncertainty and planning actions. The work suggests translating a PDDL plan into a Behavior Tree using a simple temporal network (STN). The behavior tree (BT) is a convenient framework for controlling the execution flow of the actions in the plan. Despite the high quality of such work, the uncertainty of the task's duration is not considered.

A.3.2.2 The Plan4ARI methodology

Inspired by previous works, **Plan4ARI** will tackle the problem through a "divide and conquer" approach. Specifically, we will divide the problem into five steps to ensure that the solution always grants all the problem constraints and is optimal.

²Singh, I., Blukis, V., Mousavian, A. et al. ProgPrompt: program generation for situated robot task planning using large language models. *Auton Robot* 47, 999–1012 (2023). <https://doi.org/10.1007/s10514-023-10135-3>

³Lin, K., Agia, C., Migimatsu, T. et al. Text2Motion: from natural language instructions to feasible plans. *Auton Robot* 47, 1345–1365 (2023). <https://doi.org/10.1007/s10514-023-10131-7>

⁴Garrett, C. R., Lozano-Pérez, T., & Kaelbling, L. P. (2020). PDDLStream: Integrating Symbolic Planners and Blackbox Samplers via Optimistic Adaptive Planning. *Proceedings of the International Conference on Automated Planning and Scheduling*, 30(1), 440-448. <https://doi.org/10.1609/icaps.v30i1.6739>

⁵Josh Zapf, Marco Roveri, Francisco Martin, Juan Carlos Manzanares, Constructing Behavior Trees from Temporal Plans for Robotic Applications, <https://doi.org/10.48550/arXiv.2406.17379>

- Step A) State-of-the-art tools of **Generative AI** will be used to deploy a **Knowledge Base (KB)**. The process engineer will check the KB generated architecture, which will be the cornerstone of the task planning framework. Multi-agent GenAI with Reinforcement Learning (as Claude AI 3.5) will be adopted (**starting TRL2, expected TRL4**). In other words, the creation of KB corresponds to pre-form the different functional blocks that can be used, and their properties are in a model-driven engineering platform such as MI.RA/Dexter.
 - Task planning is commonly based on the formalization of the **domain**, which consists of defining the **knowledge base**. This can be done according to different toolchains. Specifically, an expert in the process will create the KB for a class of operations by simply describing the domain using natural language. From the description, the tool will extract the domain information, such as types (objects or entities), predicates (states or conditions), and actions (operations or tasks that change states). For example, a proper KB will be created for each class of operations, such as welding applications, pick and place, quality controls, etc.
- Step B) State-of-the-art tools of **Generative AI** will be used to generate **PDDL** coherent with the **KB** from Natural Language. The idea consists of exploiting Lightweight LLM. The User, therefore, will prompt the description of the specific task, and the SW tool will turn it into a PDDL descriptor (**starting TRL4, expected TRL6**).
 - In literature, the previous step and this step are solved together. Generative AI tools generate the PDDL directly from the natural description without KB. This, however, can lead to errors in the PDDL and weakness in the description. **Plan4ARI** will exploit multi-agent generative AI that will check the PDDL according to the KB generated at the previous step. A formal verification tool will also be exploited. Practically, it does mean that the natural language used by the operator to describe the process will be turned iteratively in a PDDL until it matches the domain definition stored in the KB.
- Step C) Innovative methods will be developed to turn **PDDL into a Behavior Tree with time-variance modeling** to integrate time variability. This innovation will allow the **ARI to tackle variable temporal planning without needing any temporal planner solver (starting TRL2, expected TRL4)**.
 - At the state of the art, many solvers compute the best task plan given the time duration of the tasks. However, no temporal planner is still available off-the-shelves that also considers the variance in the duration of the tasks, and a few papers are dedicated to this issue. Since such a solver is out of the competence of the working team, we will follow an alternative path. We will translate the PDDL into a behavior tree using time variability to allow a search for the best solution over the behavior tree (see next point). This SW module will exploit STNU (Simple Temporal Network **with** Uncertainty) and consist of the straightforward extension of existing tools that turn PDDL into a Simple Temporal Network **without** Uncertainty (**starting TRL4, expected TRL6**).
- Step D) the **Task and Motion Planning Optimization** will be performed within a **Digital Twin Environment**.
 - Having a Behavior Tree that stores all the possible Task plans will make it possible to optimize the best task sequence. In the writing phase, we aim to explore the multiple parallel BT scenes using **Monte Carlo Tree Search** methodologies, allowing the identification of the global optimal plan considering the temporal variance of the actions (**starting TRL2, expected TRL4**). The digital twin will consist of turning the BT representation into the MI.RA/Dexter flowchart and generate a virtual execution of all the tasks.
- Step E) A SW module will execute the Optimal Behavior Tree and will implement Online replanning strategies according to the context state (**starting TRL4, expected TRL6**).
 - The online replanning methodologies will be deployed using heuristics. The feasibility during the execution will be the primary target since the optimization could ask for a computation time that mismatches the execution deadlines and requirements.

A.3.3 Roles and Activities Management

A.3.3.1 PI Role

The PI will coordinate the activity and be part of the team that designs the methodology, its deployment, implantation, and testing.

The full team will have a one-week touch point focused on presenting the results to the PI and the Host Institution Responsible for the MI.RA/Dexter platform.

A.3.3.2 Research Organization

The research organization will invest more than 30% of the budget. The team will be composed of a Senior Researcher and a fellow.

They will define the methodology and implement it using the first ad-hoc platform and then (from half-project to the end) inside MI.RA/Dexter.

A.3.3.3 Host Institution

The Host Institution will invest about 35% of the budget. The team will comprise the top figures responsible for the cognitive platform development, the development team, and the innovation team.

The innovation team will lead the project's goal, and it will lead the development of a use case that will verify the quality of the outcomes.

The development team will continuously interact with the Research Organization to include the scientific methodologies inside the platform. At the same time, it will continuously interact with the research organization team to share typical simulation environments and tools.

A.3.4 Expected Results and Key Performance Indexes

This module will output a unique SW platform able to translate the natural language into a part program ready to be executed. The high-level expected results are:

- Programming Time Reduction of 66%
- Programming performed by not pre-trained worker
- Management of fleet composed of 3 mobile manipulators or 4 industrial robotic arms

The performance indexes will be measured in controlled environments designed to test, while the whole system will be evaluated through a specific demonstrator.

A.3.4.1 Correctness in the Knowledge Base Generation

Different application scenarios will be tested. The autonomously generated KB will be checked by a human researcher in the field to verify the accuracy of the results. Furthermore, an automatic KB checker will be deployed.

KPIs: 10 contexts will be evaluated. | **TRL:** Such technology is TRL2 in SOTA. Our approach will increase performance; the novel approach will reach TRL4 (Fundamental Research).

A.3.4.2 Correctness in the PDDL generation

Different application scenarios will be tested, and the output will be checked by a researcher in the field

KPIs: +100 contexts will be evaluated. | **TRL:** Such technology is TRL2 in SOTA. Our approach will increase performance; the novel approach will reach TRL4 (Fundamental Research and Industrial Research).

A.3.4.3 Correctness in the BT generation

Different application scenarios will be tested, and the output will be checked by a researcher in the field

KPIs: +100 contexts will be evaluated. | **TRL:** Such technology is TRL4 in SOTA. Our approach will increase performance; the novel approach will reach TRL6 (Industrial Research).

A.3.4.4 Optimal Task Planning Evaluation

Different application scenarios will be tested, and the output will be checked by a researcher in the field

KPIs: +100 contexts will be evaluated. | **TRL:** Such technology is TRL2 in SOTA. Our approach will increase the performance, but the novel approach will reach TRL4 (Industrial Research).

A.4 Intelligent Sensing and Safe Motion Planning for Mobile Manipulators and Multi-Agents Interaction

A.4.1 Objectives

The definition of the task plan is closely related to the definition of the movement to perform it. Nowadays, the operative scene is composed of a variety of agents. Multi-mobile-manipulator teams or humans could be inside the operative scene, whether with or without fences. Such challenge scenarios led to a re-think of the definition of robot path and trajectory, considering the challenging implementation of the ISO/TS15066 cooperation rules. Such motion planning methodologies are eventually intertwined with sensing technologies to create awareness of the operative scene. This is fundamental to implementing scene semantic segmentation, symbol-to-geometry autonomous capability, and tracking all humans inside the scene.

In such a context, **Plan4ARI** will deploy a fully autonomous toolchain for motion planning in **every working condition**. In detail, **Plan4ARI** will focus on:

- **A flexible software tool to solve path planning problems in every condition.** The core of this tool will be a library that streamlines motion replanning challenges with minimal code requirements. It will serve as a valuable resource for researchers aiming to develop new algorithms while simplifying the practical application of solutions. This tool will implement a **framework to manage the offline planning and online continuous replanning**. A software architecture will be devised to oversee the execution of the robot's trajectory, continuous scene updates, and concurrent replanning.

- **We will investigate an innovative multi-path replanning algorithm.** The algorithm will be designed for fast and efficient path replanning, even when dealing with robots with numerous degrees of freedom. Instead of aiming to find an entirely new solution to reach the goal, the algorithm focuses on connecting to pre-existing paths that lead to the same goal. This approach accelerates the initial solution discovery process, which can be refined over time. Formal guarantees about the completeness and optimality of the method are provided, ensuring its reliability.
- **We will optimize the motion planning according to the time variability of uncontrollable concurrent tasks.** For example, when a human operator or moveable objects in the scene cross the robot workspace. In such a scenario, a **safety-aware cost function** will be designed to guide the replanning in generating solutions prioritizing safety compliance. The resulting algorithm will facilitate improved multi-agent cooperation, enabling the robot to efficiently adjust its motion in response to the operator's unexpected movements in unstructured tasks and environments.
- **We will conceive fast replanning strategies** that can consider all the dynamics and environmental constraints for runtime velocity modulation to improve tracking performance and react to sudden scene modifications. Lookahead micro-interpolator fails in challenging operative scenes, and innovative approaches that come from control science will be investigated and applied. Specifically, the motion planning will be designed as modules running over different computation units according to the goals (Figure 1). For example, on the one hand, the motion planner could run under the **Open IPC** where the user interface is running when the plan can be computed asynchronously (offline or online), and the computation latency is not critical. On the other hand, the motion planning problem must be solved in the **Core IPC** (Figure 1), when the interpolation must be solved in synchronously to the process execution (e.g., visual servoing)
- **Deep learning techniques for autonomous robot registration** inside the scene and **semantic segmentation** of the scene. The autonomous generation of the motion plan needs the exact localization of the robot into the scene. This will be performed autonomously by adopting a mix of hand-to-eye and eye-to-hand sensing systems. Furthermore, a robust segmentation engine will translate the symbolic description of Cartesian targets into the exact localization of such poses.
- **The integration of radar sensors to track the human inside the scene.** The objective will be to test and deploy algorithms that can exploit both 3D point clouds coming from vision systems and radar sensors. Radar sensors are the most promising technology for boosting human-robot interaction.

A.4.2 Scientific and Technical Approach

A.4.2.1 Context and State-of-the-art

Generating a new path whenever a new obstacle appears requires substantial computational resources and time. A more efficient approach involves dynamic adjustment of the existing path to quickly and efficiently respond to changes. Nevertheless, existing techniques still require significant computational power, impacting their responsiveness. Replanning algorithms need to be supported by complex software architectures. When dealing with unchanging environments, the regular approach to motion planning involves computing a path, assigning it a time parametrization, and executing the resulting trajectory. On the other hand, when it comes to motion replanning, there is a greater demand throughout the trajectory's execution. It is essential to consistently reevaluate and adjust the plan as the robot moves, smoothly transitioning to new trajectories. **This calls for an architecture that not only supervises the robot's motion execution but also maintains up-to-date scene information while simultaneously managing the replanning process.** However, in workspaces shared with humans, it is necessary to do more than ensure that the robot's path is collision-free. The robot's speed adjustments and halts are controlled by safety systems that continually monitor the workspace to guarantee human safety. The ISO/TS 15066 safety standard proposes collaborative operations like Speed and Separation Monitoring (SSM) or Power and Force Limiting (PFL) to adhere to safety regulations and protocols. The choice of strategy depends on risk analysis and aims to reduce potential hazards.

Their implementation typically limits robot speed and halts when the human and the robot are close. It is possible to reduce the idle times caused by the safety intervention by planning robot movements that minimize the safety speed limitations. Researchers have investigated this problem at various levels, proposing adaptive task planners⁶, motion planners⁷, and controllers⁸. Motion planning is pivotal in this endeavor, with researchers striving to consider the operator's presence and trigger the safety module as sparingly as possible, employing proactive or reactive approaches. **Proactive planners** pre-compute trajectories that minimize the expected interference between the operator and the robot. This relies on a model of human behavior and the minimization of collision risk, execution time, or human discomfort. While these approaches are robust to the operator's presence, duration and frequency of access, they become considerably less reliable when the operator's position is out of distribution. **Reactive planners** focus on real-time robot trajectory modifications, involving dynamic speed adjustments according to the distance from the operator and the ability to alter the path dynamically. Speed reduction works when the intrusions are short and sparse, but it is inefficient with frequent collaboration, resulting in safety stops whenever the human is nearby. An emerging trend is to

⁶S. Sandrini, M. Faroni, and N. Pedrocchi, "Learning action duration and synergy in task planning for human-robot collaboration," in 2022 IEEE 27th Int Conf on Emerging Technologies and Factory Automation (ETFA), 2022, pp. 1–6.

⁷A. Palleschi, M. Hamad, S. Abdolshah, M. Garabini, S. Haddadin, and L. Pallottino, "Fast and safe trajectory planning: Solving the cobot performance/safety trade-off in human-robot shared environments," IEEE Robot. Autom. Lett., vol. 6, pp. 5445–5452, 2021.

⁸C.-M. Huang and B. Mutlu, "Anticipatory robot control for efficient human-robot collaboration," in 2016 11th ACM/IEEE Int Conf on Human-Robot Interaction (HRI), 2016, pp. 83–90

combine strategies, employing path modifications during execution and trajectory scaling.

A promising strategy for motion planning in collaborative operations is blending reactivity with proactivity. This is achieved by integrating a reactive motion replanning algorithm with a human-aware cost function to minimize interference proactively. In this way, cooperation efficiency is improved by optimizing the trajectory for travel times and operator-induced safety slowdowns. Previous studies were limited to offline planning when considering human influence⁹, and replanning algorithms could require adaptations to handle more complex cost functions

In such a context, **sample-based algorithms, such as rapidly exploring random trees (RRT)** and **probabilistic roadmaps (PRM)**, are famous for robot trajectory planning because they can efficiently handle high-dimensional spaces. One of the critical developments in improving the efficiency of these algorithms is the concept of the **informed set**, which is particularly useful for enhancing the performance of algorithms like RRT*, an optimized version of the original RRT.

Specifically, the **informed set** is a subset of the state space that contains all the potentially be easily integrated into the replanning solutions to a given problem. In traditional sampling-based planners, samples are drawn uniformly from the entire state space, which can lead to inefficiencies, mainly when the feasible region for the optimal path is small. The informed set constrains the sampling space to focus on regions more likely to contain the optimal solution, significantly reducing the number of samples needed. For example, in RRT*, the informed set is defined as an ellipsoid that bounds the solution space. The planner converges more quickly to an optimal solution by focusing sampling within this ellipsoid. This ellipsoid is defined based on the best path cost and the distance from the start to the goal state.

The primary advantages of using an informed set in sample-based planning algorithms include **Reduced Computational Complexity**: Sampling within a constrained region (informed set) decreases the number of required samples, accelerating the convergence toward the optimal solution, **Improved Path Quality**: By focusing sampling in regions that are more likely to contain the optimal solution, informed set-based methods produce better-quality paths, with fewer iterations, **Scalability**: The use of an informed set makes these algorithms scalable to higher-dimensional spaces, which is crucial for applications, such as autonomous drones and manipulators operating in complex environments.

The availability of fast planning methodologies based on sample-based approaches is decisive for the path design, but it is hard to integrate the dynamics constraints of the robot. Dynamics computation would insert an excessive workload in the state exploration and exploitation.

A different approach uses **Model Predictive Control (MPC)** as an active filter to generate a robot controller's path or trajectory reference. MPC is used for trajectory generation rather than feedback control in this configuration. **MPC** can be designed to calculate a **sequence of future reference points** for the robot to follow based on the desired objectives (such as reaching a target location while avoiding obstacles). MPC solves an optimization problem over a prediction horizon, considering system dynamics, constraints (like robot speed, acceleration, and workspace boundaries), and external factors (such as obstacle positions or dynamic environments). In this context, the MPC output is a smooth, feasible trajectory that serves as the **reference input** for the lower-level controller.

The MPC's output is smoother and respects the robot's physical constraints, ensuring the path is feasible. This trajectory can be updated in real-time based on feedback, making it more adaptive to environmental changes or disturbances. There are multiple key benefits of using MPC for trajectory generation. **Constraint Handling**: MPC allows you to account for physical constraints like maximum velocity, acceleration, and obstacle avoidance directly in the trajectory generation process. **Adaptivity**: The real-time nature of MPC allows for trajectory adjustments in dynamic environments or under disturbances. **Smooth and Feasible Trajectories**: MPC naturally generates smooth, feasible trajectories that meet the system's limitations. Model Predictive Control (MPC) has significantly contributed to the trajectory generation for autonomous robots, with applications ranging from ground vehicles to aerial drones.

One relevant study focuses on MPC-based trajectory generation for aerial robots¹⁰, where the MPC algorithm dynamically adjusts paths while considering environmental uncertainties. This approach uses learned penalization functions within the MPC framework to avoid redundant coverage of already explored areas, optimize the trajectory for search and coverage missions, and make it ideal for aerial robotic systems. Another recent paper presents **SOMTP**¹¹, a learning-based optimization framework integrated with MPC for safe trajectory planning in robotics. It uses Control Barrier Functions (CBFs) to ensure safety around obstacles. The combination of MPC and CBF allows robots to plan dynamically while respecting real-time constraints, particularly in highly nonlinear, non-convex environments. Furthermore, many MPC-based frameworks have been applied for autonomous vehicle path tracking, enhancing real-time decision-making and path adjustments under dynamic conditions like obstacle avoidance and lane changes¹².

⁹R. Laha, W. Wu, R. Sun, N. Mansfeld, L. F. Figueredo, and S. Haddadin, "S*: On safe and time efficient robot motion planning," in 2023 IEEE Int. Conf. Rob. Aut. (ICRA), 2023, pp. 12 758–12 764

¹⁰T. Power and D. Berenson, "Learning a Generalizable Trajectory Sampling Distribution for Model Predictive Control," in *IEEE Transactions on Robotics*, vol. 40, pp. 2111–2127, 2024, doi: 10.1109/TRO.2024.3370026.

¹¹Liu, Yifan, You Wang, and Guang Li. "SOMTP: Self-Supervised Learning-Based Optimizer for MPC-Based Safe Trajectory Planning Problems in Robotics." arXiv preprint arXiv:2405.09212 (2024).

¹²Johny, D., Jisha, V.R. (2023). Model Predictive Control-Based Trajectory Generation and Tracking of an Electric Vehicle. In: Siano, P., Williamson, S., Beevi, S. (eds) Intelligent Solutions for Smart Grids and Smart Cities. IPECS 2022. Lecture Notes in Electrical Engineering, vol 1022. Springer, Singapore. https://doi.org/10.1007/978-981-99-0915-5_22

A further research challenge consists of tracking and segmenting all the information in the scene. This point is crucial to:

- Allow **fully symbolic programming**. In other words, the user indicates only a symbolic target to the robot, while the robot autonomously converts in Cartesian/Joint targets.
- Tracking the moveable objects in the scene is the boost to enable online replanning and – therefore – an improvement of the efficiency of the systems.

Sensing systems based on 3D cameras have achieved high performance in many conditions. The segmentation of the images is impressively improving every year, and FOSS algorithms are available. The main issue related to 3D camera systems is related to the fact that they are slow to track fast movement (like human movements), and only one camera in the market is certifiable safe – to the best knowledge of the PI.

In automotive, it has been more than a decade since the fusion of radar sensor and camera systems allowed impressive performance and an intrinsic level of safety. On the one hand, radars are extremely fast, allowing the updating of the scene status. On the other hand, the data's resolution and noise are significantly worse than the 3D images.

The fusion of camera and radar sensors is highly appealing to robotics. The most relevant trend considers **Deep Learning Dominance**, i.e., moving towards deep learning-based fusion, where networks can learn complex relationships between the sensors. New architectures like transformers are being explored for better sensor integration. These methods are also investigated to achieve high performance in **sensor calibration and alignment**, ensuring precise spatial alignment between radar and cameras, and are also advancing. Remarkably, radar and camera data fusion is evolving rapidly with AI and machine learning advancements, leading to more accurate, reliable, and safe autonomous systems.

A.4.2.2 The Plan4ARI methodology

Being inspired by my previous works in the field and exploiting my FOSS software (<https://github.com/JRL-CARI-CNR-UNIBS/OpenMORE>), **Plan4ARI** will tackle the problem through an innovative framework designed explicitly for motion planning and re-planning. Precisely, the technical and scientific approach will follow

- Development of an SW module for the **sensing fusion**. The fusion of images and radar sensing systems will be addressed to have a high-frequency 3D point cloud of the cell.
 - The goal is to boost the update frequency of the scene by a double sensing layer. The internal fast layer will comprise radar sensing technologies that create low-resolution and noisy point clouds at high frequencies. The outer layer will be composed of 3D camera systems that update the scene state at low frequency, over which it is possible to perform accurate registration and segmentation. This framework will allow the high-frequency tracking of moveable objects in the scene.
 - The module will integrate state-of-the-art fusion and segmentation techniques to extract information from 3D colored point clouds. This will allow the robot to interface with the object database and program the robot using symbols (e.g., descriptors of the objects in the scene) and not Cartesian goals. If the system fails in object identification, the Cartesian goals will be inserted with an easy-to-tune user Cartesian reference frame.
- Development of a tool for advanced understanding of the scene.
 - First, thanks to the sensing system's high-rate enriched sensing fusion, it will be possible to implement advanced state-of-the-art registration methodologies. **This will allow continuously updating the robot's position in the scene.** Like in mobile manipulators, the high localization will allow to tackle straightforwardly the semantic segmentation
 - Module A3 will allow the creation of the KB of each scenario easily. This powerful tool will enable the implementation of advanced semantic segmentation techniques. The segmentation and pose estimation of the objects of the scene will allow for the symbolic planning of the robot. No Cartesian poses will be necessary anymore when programming the Robot.
- Development of adaptive strategies across task planning and motion planning to minimize the need for safety interventions – that means to reduce wasted time and enhance collaborative productivity. **Building upon the existing safety module**, we will deploy sample-based strategies that **proactively** and **reactively** oversee the robot's actions, ensuring smooth cooperation with the operator in shared spaces. I aim to bridge the gap between safety-aware proactive and reactive methods.
 - The conventional approach to path replanning needs will be reconsidered. Ensuring a collision-free trajectory does not necessarily translate to a safety-compliant robot's behavior. Modifying the path to avoid collisions might cause the safety system to intervene significantly, resulting in slowdowns or stops, which could undermine the robot's effectiveness. As a result, replanning algorithms need to be aware of the effects of the safety module. These algorithms will consider humans as obstacles and factors that affect how the trajectory should be modified to achieve efficient behavior
 - The approach I will pursue will integrate a safety-aware cost function based on ISO/TS 15066 into a fast-replanning algorithm, changing the robot's path in runtime. The primary objective is to minimize the execution time of the trajectory, considering the estimated speed limitations. The method will address the limitations of proactive approaches by allowing online adaptation of the plan to accommodate unexpected movements of the operator. It also improves the efficiency of reactive approaches, going beyond simple collision-free path replanning and actively seeking safety-aware

solutions, aim to prevent hazardous situations instead of reacting to it. For this purpose, we draw inspiration from my work¹³ to introduce a safety-aware cost function based on ISO/TS 15066 that can be easily integrated into the replanning algorithm.

- To overcome the computational complexity, which can impact robot promptness, I will adopt a **sampling-based approach**. This method seeks solutions by randomly sampling the robot search space, and it is favored for its simplicity, adaptability, and ability to scale, even when dealing with large search spaces (e.g., robots with many degrees of freedom).
- The idea consists of dramatically improving the performance of RRT-like sampling-based path planners by combining **admissible informed sampling and local sampling** (i.e., sampling the neighborhood of the current solution). **An adaptive strategy** will regulate the trade-off between exploration (admissible informed sampling) and exploitation (local sampling) based on online rewards from previous samples.
- Development of a **Model Predictive Controller** for path planning running alongside the adaptive sampling-based motion planner. This will integrate the controlled model of the robot to generate the best trajectory concurrently with the robot execution task, granting the prescribed performance.
 - The MPC is renowned for its performance in optimizing the control law and non-linear complex systems. **Plan4ARI** will focus on online speed adaptation to reduce the tracking error when the dynamics load is essential for the actuators. Alternatively, the robot will exploit the residual degrees of freedom along a path to reduce the jerk of the trajectory. For example, suppose the robot end-effector can rotate freely along the tool axis (e.g., in many welding applications, in machining, and in painting, it happens frequently). In that case, the MPC will compute the best orientation to reduce the maximum/mean torque for the motors along the required time window.

A.4.3 Roles and Activities Management

A.4.3.1 PI Role

The PI will coordinate the activity and be part of the team that designs the methodology, its deployment, implantation, and testing.

The team will have a one-week touch point focused on presenting the results to the PI, the Host Institution Responsible for the MI.RA/Dexter platform and the Host Institution are Responsible for the low-level control.

A.4.3.2 Research Organization

The research organization will invest about 35% of the budget in this topic. The team will comprise two fellow researchers directly, followed by the PI.

They will define and implement the methodology using the ad-hoc platform and then (from half-project to the end) inside MI.RA/Dexter (for the offline/online motion planner) and the **Core IPC** (for the Model predictive controller).

A.4.3.3 Host Institution

The Host Institution will invest about 35% of the budget to integrate. Different top figures will be engaged: the responsible for the cognitive platform development, the responsible for the low-level controller development, and the responsible department of the vision systems.

Three working teams will be engaged. First, a working team will mainly focus on developing innovative methodologies for scene update, registration, and semantic segmentation. The second working team will integrate the offline/online motion planner with the MI.RA/Dexter platform. The second working team will put effort into deploying the Core IPC controller with a double goal. First, the MPC must be integrated into the Core IPC. Second, to create a powerful async communication channel with the MI.RA/Dexter platform for safe continuous streaming of reference trajectories for the robot.

Both teams will continuously interact with the Research Organization to include the scientific methodologies inside the platform. At the same time, it will continuously interact with the research organization team to share common simulation environments and tools.

A.4.4 Expected Results and Key Performance Indexes

This module will output a unique SW platform able to translate the natural language into part program ready to be executed. The high-level expected results are:

- Fully autonomous generation of the motion plan in all the required working conditions. Both joint and Cartesian planning will be supported.
- Reduction of the execution time for interaction tasks by 30% to state-of-the-art motion planners.
- Safe adaptation of the velocity according to any working conditions

¹³M. Faroni, M. Beschi, and N. Pedrocchi, "Safety-aware time-optimal motion planning with uncertain human state estimation," *IEEE Robot. Autom. Lett.*, vol. 7, no. 4, pp. 12 219–12 226, 2022.

The performance indexes will be measured in controlled environments designed to test, while the whole system will be evaluated through a specific demonstrator.

A.4.4.1 Computation Time Target

Plan4ARI will test at least 50 application scenarios, and the output will be checked by benchmarking it with OMPL planners (<https://ompl.kavrakilab.org/index.html>). As a note, the planners running on GPU will not be compared.

KPIs: Computation time, computed as mean over +100 contexts. | **TRL:** Such technology is TRL5 in SOTA. Our approach will increase performance, but the novel approach will reach TRL7 (Industrial Research).

A.4.4.2 Scene Update Target

Plan4ARI will test at least 50 Different application scenarios. The sensor fusion of radar technologies and 3D images will be tested over the standard 3D vision-based systems.

KPIs: The accuracy in the registration in +100 contexts will be evaluated. The fused information is expected to grant a better accuracy of about 20% using standard registration policy (e.g., IPC) | **TRL:** Such technology is TRL4 in SOTA. Our approach will increase performance, and the novel approach will reach TRL6 (Fundamental Research).

A.4.4.3 Trajectory adaptation to reduce dynamics load

Plan4ARI will test at least 50 Different application scenarios.

KPIs: The tracking performance will be increased by about 50% | **TRL:** Such technology is TRL5 in SOTA. Our approach will increase performance, and the novel approach will reach TRL7 (Industrial Research).

A.5 Advanced Robot Motion Controlling through Machine Learning Methods

A.5.1 Objectives

The inner layer of the **ARI** (Figure 1) is the high-rate real-time motion controller. In such a context, the system identification and control law design are interweaved, and edge generative AI technologies are scarce in challenging domains like robotics, where **high non-linearity poses significant challenges**. Recent pioneering approaches tackle the new generation of AI-driven motion controllers by learning meta-dynamical models of a high-dimensional physical system to be integrated with Deep Model Predictive Control frameworks in robotics. Others integrate lightweight networks inside the motion control law.

Significant drawbacks are, however, still blocking the adoption of advanced learning-based control strategies in industrial robot controllers:

- The computational burden
- The robustness and adaptability
- The integration with the robot sensing (e.g., vision and force sensors) and their performance

Plan4ARI aims to be disruptive in conceiving and deploying a new generation of motion controllers that put the adaptivity to variable scene conditions at the center - i.e., that preserves safety and controllability in every condition.

- **To innovate in deploying controllers that guarantee performance even under actuator faults and motion constraints.** The goal consists of innovating the tracking control problem of robotic systems with guaranteed transient behavior and steady-state performance in the presence of modeling uncertainties, unexpected actuation failures, and motion constraints.
- **To innovate in deploying controllers considering elasticity inside the industrial robot joints.** The goal is to innovate robotic systems' tracking control problem when the joints' elastic effect is not neglectable.
- **To innovate in the robustness and reliability of neural controllers.** Given the hard real-time and control robustness requirements, AI applied to robot control is challenging. Plan4ARI will boost a new generation of adaptive neural controllers designed with **learning policies that can run over lightweight industrial PCs** - i.e., the most powerful CPUs will never control industrial robots.

A.5.2 Scientific and Technical Approach

A.5.2.1 Context and State-of-the-art

A common trend in industrial robotics is to couple full-state observers with fine-tuned axis controllers, where the identification of the robot dynamics plays a fundamental role.

This classical approach relies on important assumptions: it allows a complete physical understanding of the control law, debugging, and the fact that the robot model can be fully deployed. However, the elastic modeling of the robot introduces severe limitations, given the high order and the relative modeling complexity.

The literature reports an incredible number of possible full-state observers to estimate the complete state of a system (e.g., positions, velocities, accelerations) when some parts of the state are not directly measurable. Among the others, the **Unscented**

Kalman Filter¹⁴ (UKF) avoids linearization by using an unscented transform to propagate mean and covariance through the system's nonlinear dynamics. This provides better estimates than the Extended Kalman Filter (EKF) for highly nonlinear systems. It is used in applications with more pronounced nonlinearities, such as in high-speed robotic navigation, UAVs, or in cases with significant sensor noise and complex dynamics. However, it is computationally expensive and poorly manageable for real-time applications that run on IPCs. Another observer that is very commonly used is the **sliding mode observer**. This is a robust observer used primarily for systems with uncertainties or disturbances. The sliding mode observer uses a discontinuous control signal to "slide" the estimated state to the true state in the presence of disturbances. It is particularly useful in systems with bounded noise and disturbances. Sliding mode observers are applied in robotic manipulators and mobile robots, especially for estimating states in the presence of model uncertainties or external disturbances. Although it is highly robust to disturbances and model inaccuracies, it can introduce chattering due to discontinuous control, though techniques exist to mitigate this. Often, it is used for force estimation in robotic manipulators or fault detection in industrial robots.

A recent trend is the adoption of a **High-Gain Observer** that grants fast and accurate state estimation (e.g., joint velocities or forces). However, they are extremely sensitive to noise, and model inaccuracies can be problematic when using high feedback gains.

An alternative approach to having a high-performing observer is to have a full neural controller. Neural controllers¹⁵ represent a frontier for industrial robot control because they offer many potential advantages over traditional control strategies. These advantages are rooted in the capabilities of neural networks to handle complex, nonlinear, and dynamic systems, which are common in industrial robotics. The most interesting properties for neural controller are:

- **Learning and Adaptability.** Traditional controllers, such as PID (Proportional-Integral-Derivative) or model-based controllers, are often tuned for specific tasks and environments. They struggle when operating in dynamic, uncertain, or unstructured environments that are becoming more common in modern industries. The Neural Controllers show the advantage that can learn from data, adapting their behavior as tasks, conditions, or environments change. This ability to learn and generalize from experience allows robots to handle a wider variety of tasks without needing to reprogram or fine-tune control parameters.
- **Handling Nonlinearities and Complex Dynamics.** Industrial robots often deal with highly nonlinear systems, such as joint friction, backlash, and dynamic loads, which can be difficult to model and control using conventional methods. The Neural Controllers show the advantage that neural networks excel at approximating complex, nonlinear relationships without the need for detailed system models. This enables better control of robots with complex dynamics.
- **Real-Time Adaptation to Unknown Disturbances.** Industrial environments are often unpredictable, with variable external forces, or sudden changes in load. Conventional controllers require precise models and struggle with real-time adaptation to such disturbances. The Neural Controllers, particularly those integrated with reinforcement learning (RL) or adaptive control methods, can adjust in real time to unforeseen disturbances, improving the robot's ability to maintain stability and performance even in uncertain conditions.
- **Data-Driven Control.** In traditional control systems, deriving accurate models of robot dynamics or task environments requires time-consuming analytical work. Data-driven control simplifies this by learning directly from operational data, making the process faster and more adaptable to changes. The Neural Controllers show the advantage that they use data to "learn" system behavior without the need for explicit modeling, offering an efficient way to create control policies. This is particularly useful when the robot's environment is too complex or unknown for model-based control.
- **Improvements in Sensing and Perception.** Robots are increasingly equipped with advanced sensors, including vision systems, force-torque sensors, and haptic feedback devices. Traditional controllers often have difficulty integrating and making real-time decisions based on this flood of sensory data. The Neural Controllers are highly effective at processing large amounts of sensor data, such as camera inputs, LiDAR, or touch sensors. This ability enables robots to adjust their behavior based on rich sensory input.
- **Task Generalization and Autonomous Learning.** In many industrial settings, robots are required to perform multiple tasks or variations of the same task. Conventional controllers usually need to be fine-tuned for each task, which limits their flexibility. The Neural Controllers, particularly those trained through reinforcement learning, can generalize across tasks. They can learn from a broad set of experiences and apply those learnings to new, unseen tasks without the need for manual reprogramming.
- **Cost Efficiency and Flexibility.** Traditional robot control systems often require expensive hardware and fine-tuned software for specific applications, which can be costly and time-consuming to develop and maintain. Neural controllers can reduce development time and costs by learning directly from data and reducing the need for complex, hand-crafted control logic. As neural networks evolve, they could also allow for more general-purpose robots to handle a broader range of tasks with fewer customizations.

Recently, some kinds of neural network (NN) control methods, such as NN-based inverse control, indirect adaptive NN control, and direct adaptive NN control, have been proposed to deal with dynamic uncertainties. Several elegant adaptive neural controls

¹⁴ Siciliano, B., Sciavicco, L., Villani, L., & Oriolo, G. (2010). *Robotics: Modelling, Planning and Control*. Springer Science & Business Media.

¹⁵ aa

(ANCs) have been presented for nonlinear systems to simplify controller design and avoid model estimation. However, the given predefined performance requirements for the convergence rate, maximum overshoot, and steady-state error have not been studied fully.

A neural learning control with pre-scribed performance was presented for a class of SISO uncertain nonlinear systems. NNs are generally introduced to deal with complex nonlinear systems owing to their universal approximation property. Still, most of the existing literature has not fully addressed NN learning abilities for unknown system dynamics.

Therefore, even for the same control task, the NN weights must still be retrained, wasting unnecessary energy and time. This differs from human learning behavior: "learning by doing" and "doing with learned experience." An ANC should be able to utilize the learned knowledge for a similar task. To obtain accurate convergence of the parameter estimates to optimal constant weights, the **persistent excitation (PE)** condition of internal signals in the closed-loop system is usually required to be satisfied. However, the satisfaction with PE conditions is a very challenging task. More recently, it was shown that a partial PE condition of the regression sub-vectors along recurrent trajectories can be guaranteed by localized radial basis function (RBF) networks.

The fundamental challenge for neural controllers consists of robustness, which is not easy to demonstrate, and the behavior could be challenging to test under every working condition.

To overcome such limitations, evaluating the integration of neural controllers with Prescribed Performance Controllers (PPCs) could be interesting. In contrast to traditional control strategies that may react to deviations or disturbances as they occur, PPC predefines the desired performance and drives the system towards that target throughout its operation. The key feature of prescribed performance control consists of defining predefined performance bounds. The desired performance, such as error convergence rates, bounds on transient behavior, and steady-state accuracy, is set in advance. This provides a precise performance goal from the beginning. PPC typically transforms the system's error dynamics into a new form, ensuring these errors evolve according to prescribed limits. For example, the control system may ensure errors decrease within specified bounds or rates over time. PPC is robust to system uncertainties, external disturbances, and nonlinearities, ensuring that the system meets the performance criteria despite unpredictability.

A.5.2.2 The Plan4ARI methodology

Plan4ARI will approach the problem by investigating the adoption of Prescribed Performance Constraint methodologies integrated with Learning Methodologies (radial basis function networks RBFNN) to implement **fast MIMO controllers that guarantee performance even under actuator faults and motion constraints**.

Plan4ARI conceives a dynamic learning scheme for an industrial robot manipulator that considers the elastic components with prescribed performance.

- First, ANC will be investigated to **guarantee prescribed performance requirements** and achieve knowledge acquisition and storage of unknown system dynamics.
- The stored knowledge will be reused to design **a neural learning controller** to improve control performance.

Specifically, **Plan4ARI** will combine the constrained tracking control performance and the predefined performance functions. We will conceive an innovative error transformation method to transform the original constrained tracking control problem into an equivalent unconstrained stabilization variable. A filtering tracking error will be deployed on the transformed errors for the unconstrained control variable. Then, an **adaptive neural controller** will be designed to guarantee that the tracking errors converge to a small neighborhood of zero while satisfying the prescribed performance. The state transformation will be employed to transform the closed-loop systems into linear time-varying (LTV) systems with very small perturbations, which achieves the accurate convergence of neural weight estimates using the partial PE condition of RBF NN. Subsequently, the converged neural weights will be expressed and stored as constant values. Using the stored knowledge, a neural learning control scheme with prescribed performance will improve the tracking performance for the same or similar control task.

The PROs of the proposed approach are:

- Easy integration of compliance modeling
- The full-state observer is not necessary
- The MIMO does not need any matrix inversion
- The integrator term is straightforwardly modelled
- The computation time should be limited

A.5.3 Roles and Activities Management

A.5.3.1 PI Role

The PI will coordinate the activity, and he will be part of the team for the design of the methodology, its deployment, implantation and test.

The full team will have a one-week touch point, focused on presenting the results to the PI, the Host Institution Responsible for the low-level control.

A.5.3.2 Research Organization

The research organization will invest on this topic about the 25% of the budget. The team will be composed of two fellow researchers directly followed by the PI.

They will define the methodology and implement it using first ad-hoc platform and then (from half-project to the end) inside the **Core IPC**.

A.5.3.3 Host Institution

The Host Institution will invest about 27.5% of the budget to integrate. Different top figures will be engaged: the responsible of the low-level controller development, and the responsible of department of the vision systems.

Two working teams will be engaged. First, a working team will mainly focus on supporting the research organization in developing a realistic digital twin of COMAU robots that can be exploited in the development phase. Furthermore, such a team will support the research organization in defining the evaluation metrics, designing the experiments, and measuring the performance.

The second working team will put effort into integrating the control strategies inside a prototype of a robot controller.

Both teams will continuously interact with the Research Organization to share common simulation environments and tools.

A.5.4 Expected Results and Key Performance Indexes

The motion controller will consist of a running SW module. The expected performance is

- Fully autonomous auto-tuning of the dynamics and control parameters.
- Reduction of about 10% of the mean tracking error
- Reduction of about 30% of the overshoot peak in high-dynamics requests

The performance indexes will be measured in controlled environments designed to test, while the whole system will be evaluated through a specific demonstrator.

A.5.4.1 Computation Time Target

At least +200 application scenarios will be tested to stress the computation time of the module.

KPIs: Computation time, computed as mean over +200 contexts. | **TRL:** Such technology is TRL2 in SOTA. Our approach will increase the performance, but the novel approach will reach TRL5 (Industrial Research).

A.5.4.2 Deployment Time Target

At least +200 application scenarios will be tested to stress the computation time of the module.

KPIs: The deployment time of a running robot controller relies mainly in the dynamic's parameter and control law gain definition. The target is to reduce the deployment time of a 50% to the actual COMAU time | **TRL:** Such technology is TRL4 in SOTA. Our approach will increase the performance, and the novel approach will reach TRL6 (Industrial Research).

A.5.4.3 Tracking performance

At least +200 application scenarios will be tested to stress the computation time of the module.

KPIs: The mean tracking error will be reduced of about 10%. The overshoot tracking error will be reduced of about 30% | **TRL:** Such technology is TRL5 in SOTA. Our approach will increase performance, and the novel approach will reach TRL7 (Industrial Research).

A.6 Trustworthiness and Human-Centric Optimization

A.6.1 Objectives

Plan4ARI aims to be disruptive in assessing the interaction between human operators and AI-powered robots. **Plan4ARI** will advance the SOTA in subjective and objective measurement tools, putting human operators at the center of the holistic control framework. **Plan4ARI** to build (i) a new scale for the AI-powered human-robot collaboration, (ii) its evaluation through broad questionnaire evaluation (outside consortium), and (iii) its integration in the control framework to support the decision-making platform. Specifically, **Plan4ARI's** ambitions are multifaceted.

- **To innovate the recruitment criteria.** Common and working knowledge defines the human approach to technology. Trust is grounded in the perception of safety and the knowledge of regulations: operators must trust a large set of hardware and software tools, which often need to be more transparent to users.
- **To investigate the criteria that define the perceived quality of working** with an AI-powered robotic companion. This refers to how the collaborative robot and operator accomplish the job effectively. **Plan4ARI** will investigate the relationship between operators and AI-powered robots in terms of feelings and relevance.
- **To measure the mental/physical stress** before, during, and after shifts with AI-powered collaborative robots.

- **To overcome dyadic interactions modeling**, measure the interplay of different actors into account. Elements include engagement and dynamic intention recognition/space negotiation in the shared work environment.
- **To contextualize the evaluation metrics on the manufacturing practices**, which are often locally negotiated. In other words, create flexible and adaptable metrics for specific work environments.
- **To include the temporal dimension of the interaction**, exploiting data-driven evaluation metrics that allow for continuous monitoring of the interaction quality during the task execution.

A.6.2 Scientific and Technical Approach

Plan4ARI will create tools for evaluating AI-powered collaboration, using subjective and objective evaluation and even advanced sensing when needed. The development requires close collaboration between SSH and CS/ENG to define relevant factors of interaction quality and map them to measurements. **Plan4ARI** research on trust and usability assumes that these properties unfold dynamically and evolve. The efficiency of the process, the time needed to complete the task, quality problems arising during the operation, and sudden/unexpected breaks in the interaction will indicate the interaction's status. Once this mapping between the objective input data and the perception of trust has been established, it will allow the continuous monitoring of the complete process. Subjective and Objective assessment tools will lead the modeling of human behavior inside the KB. Such tools will be questionnaires, and their design will follow psychology's most consolidated scientific practice. The "trustworthiness loop" will be assessed by Consortium workers due to the low level of expected TRL for the **Plan4ARI** technologies. **Plan4ARI** will precisely measure the system's performance and human acceptance by including or not including objective measurement tools within the loop of controllers, i.e., continuously updating the KB.

A few works¹⁶ have been designed to evaluate trust in industrial HRC. Still, they often use standard small payload industrial robots and no collaborative robots. Many SOTA works¹⁷ that assess trustiness must be more closely related to real-world usage. Conversely, works unrelated to robotics applications¹⁸ are robust scales for assessing a task's mental and physical workload. Old works¹⁹ presented a scale for assessing the trust of operators, but they used social robots, and recently (sub)scales for assessing "fluency" in HRC²⁰ have been proposed. While presenting a benchmark and ground truth for the trustworthiness of a system, questionnaires provide a subjective and momentary evaluation. Updating an agent's mutual trust evaluation is indispensable for a dynamic AI-powered environment. To this end, **objective measures** are necessary. Trust measurement combines robot and human performance, where the real-time trust measurement of the human is based on hand speed during part manipulation and handover tasks²¹. Another approach relates real-time sensing of physiological signals and EEG to human-machine trust²². The correlation between cognitive load and trust can be exploited using EDA measures for trust assessment during semi-automated robot operation²³. Others use body-worn inertial sensors to detect distrust. The University of Alborg recently presented the first benchmark data set for trust assessment from motion data²⁴. As a final consideration, five properties of a trustworthy AI-powered robot are remarkable: (1) safe in any uncertain/ dynamic environment; (2) secure; (3) healthy and fault tolerant; (4) trusted and easy to use to enable; (5) compliant with the law and ethical expectations. Still, AI-powered Robots have not yet reached this stage, and new acceptance models must ensure both worthiness and trustiness, and these models will be the **Plan4ARI** outcome.

Plan4ARI's methodology in SSH consists of:

- **Two-stage design of the subject evaluation forms.** First, a draft evaluation will be designed according to the context, the human factors, and the psychological and ethical aspects. Then, a first evaluation group of about 50 workers will be identified. Then, a second evaluation form will be defined according to a statistical analysis of the gathered data. Such an evaluation form will then be submitted to workers (+100) to build a dataset on the trustiness of AI-powered HRC. Eventually, the evaluation form will be made available online to expand the possible applicants.
- **Develop a data-driven methodology to objectively assess the human acceptance** of AI-powered collaborative robots, considering the context. First, such metrics will rely on continuous interaction quality monitoring during task execution. Second, the monitoring will be as non-intrusive as possible to reduce the bias introduced by wearing the operators with sensors and overloading them with further distraction elements.
- **Modelling SSH into the KB**, the core of the **Plan4ARI** digital representation.

¹⁶George Charalambous, et al.; "The Development of a Scale to Evaluate Trust in Industrial Human-robot Collaboration"; In IJSR, Vol. 8, 2016.

¹⁷Kristin Schaefer; "The Perception and Measurement of Human-robot Trust"; PhD dissertation, University of Central Florida, 2013.

¹⁸Sandra G. Hart et al. "Development of NASA-TLX (Task Load Index): Results of Empirical and Theoretical Research"; In: A.P., 52, 1988.

¹⁹Bonnie M. Muir; "Trust in automation: Part I. Theoretical issues in the study of trust and human intervention in automated systems"; In Erg., 37(11), 1994.

²⁰Guy Hoffman; "Evaluating Fluency in Human-Robot Collaboration"; In: IEEE Transactions on Human-Machine Systems, Vol. 49 (3), 2019, Pages 209 – 218.

²¹S. M. Rahman, et al. "Trust-based compliant robot-human handovers of payloads in collaborative assembly in flexible manufacturing," in 2016 IEEE CASE.

²²W.-L. Hu, K. Akash, N. Jain, and T. Reid, "Real-time sensing of trust in human-machine interactions," IFAC-PapersOnLine, vol. 49(32), pp. 48 – 53, 2016.

²³M. Lochner et al. "Trust and cognitive load in semi-automated UAV operation," in Proc 31st Australian Conf on Human-Computer-Interaction., 2020.

²⁴Rehm, M., Hald, K., & Pontikis, I. Benchmark Movement Data Set for Trust Assessment in Human-Robot Collaboration. In *Proc. of HRI 2024* IEEE.

- **Integration** of the continuous **metrics** and the **subject evaluation inside the holistic control architecture** and the dynamic risk assessment (through a proper definition of the **KB**, its continuous update, and the exploitation of such information by the controllers).

Plan4ARI will follow a human-centric design approach, allowing us to answer the questions on how to design the role of humans, considering new dynamics with safe, intelligent devices, intelligent systems, and intelligent automation. The SSH will include Ethics and liability. **Plan4ARI** will study responsibility and answerability, and thus accountability as the acknowledgment and reporting of any potential negative implications of robotics system adoption, especially in domains where system reliability is crucial. These aspects will be one of the pillars of the questionnaire design. The questionnaires, indeed, will measure all the levels of human-robot interaction.

Plan4ARI will be fully compliant with **data privacy and management**. The consortium will put significant effort into identifying each party's obligations to use and protect personal data and apportion risk.

A.6.3 Roles and Activities Management

A.6.3.1 PI Role

The PI will coordinate the activity, and he will be part of the team for the design of the methodology, its deployment, implantation and test.

The full team will have a one-week touch point, focused on presenting the results to the PI, the Host Institution Responsible for the Innovation.

A.6.3.2 Research Organization

The research organization will invest on this topic about the 10% of the budget. The team will be composed of one Senior Researcher, that will have in charge all the methodology definition, and the design of the experiments.

A.6.3.3 Host Institution

The Host Institution will invest about the 2.5% of the budget to integrate, i.e., 100k€. The responsible of department of innovation will be engaged.

The working team will assist the research organization in providing significant use case to be modeled, support the modeling of the robot's usage, identifying the client to be interviewed, supporting the interview, and analyze the results.

A.6.4 Expected Results and Key Performance Indexes

A.6.4.1 Subjective Metrics for Trustworthiness Assessment – System Feature

Development of new subjective metrics for trustworthiness evaluation, respecting ethical, legal, and liability aspects. **KRMs:** +100 workers will be engaged. Statistics on trustworthiness will be made available as an open dataset. | **TRL:** Such technology is TRL9 in SOTA but has insufficient performance. Our approach will increase performance, but the novel approach will reach TRL5.

Plan4ARI will design a new set of questionnaires on AI-powered collaborative robots according to the best practices in applied psychological science²⁵.

A.6.4.2 Objective Metrics for Trustworthiness Assessment – System Feature

Developing new objective metrics for the trustworthiness evaluation of the AI-powered human-robot collaboration, respecting ethical, legal, and liability aspects.

KPIs: +10 workers within the Consortium will be engaged. Statistics will be made available as an open dataset. | **TRL:** Such technology is TRL3 in SOTA. Our approach will advance it to TRL4.

The new objective metrics will overcome the standard dyadic interactions modeling and consider the context; they will be flexible and adaptable to the dynamic environment. The metrics will be data-driven to allow for continuous monitoring of the interaction quality during the task execution.

B Description of the participating subjects

B.1 Principal Investigator

B.1.1 Scientific Contributions

I have been in the research field for more than 15 years, and my scientific publications are enclosed in my CV. I am an Associate Editor for RCIM the most prestigious journal paper in the field of applied robotics science, edited from Elsevier. I have about ten patents deposited, and one international patent has been assigned to a company and will be active until 2037.

²⁵The Institute for Cognitive Sciences and Technologies of CNR will provide the SSH scientists

I coordinated or was responsible for 17 projects, of which 8 were EU projects (FP7, H2020, HE), 2 EU meta-projects, one UK collaboration, one national, two regionals, and 3 Research and Innovation Contracts with Companies. The 18 projects I managed had a financial capacity of just over €6M (max €810k, min €15k). I participated as a unit member in 12 projects (WP Leader and/or Task Leader).

My scientific production has focused on developing methodologies to extend the “capabilities” of industrial robots so they can be used in unstructured environments, significantly increasing the added value of the production system and improving the quality of work for human operators. My contributions have focused on three enabling technologies for industrial robots (only the most essential products are listed): motion and task planning, interaction control, modeling, and robotized machining.

In addition to contributions in these areas, I investigated the use of anthropomorphic robots for rehabilitation and the development of new mechatronic devices and new robots for cranial or spinal surgery [AJ44]. Another area where I sought to make original contributions was the development of systems to make industrial robotic applications safe, even without barriers. Specifically, we investigated how certified and non-certified sensor networks could be used. This activity is characterized by a multidisciplinary solid involving the CNR-IEIT institute.

Some works have been more related to new frameworks, and only recently have I started investigations into the use of Deep and Reinforcement Learning in the industrial sector. The achievement of the scientific results described is linked to both scientific and managerial methodological skills.

Regarding methodology, the results demonstrate that I possess (i) scientific solid rigor, (ii) analytical and synthetic skills in multi-body system modeling, (iii) a propensity for multidisciplinary, (iv) rigorous study of state-of-the-art

Regarding managerial skills, the results show that I possess (i) excellent project management skills, (ii) strong networking and lobbying capabilities, (iii) administrative and managerial skills acquired through the management of European projects, (iv) attention to the development of the human resources I manage. These managerial skills have been fundamental for me in managing three laboratories in parallel and four PhDs. The two aspects on which I have focused my attention have been (1) very detailed activity planning and (2) formalization of knowledge within the labs, explicitly defining methods for developing software and mechatronic projects that enable sharing and maintenance.

B.1.2 Technological Transfer

The transfer and application of my research results for the benefit of the scientific community over the years have followed three operational models:

- (A) participation in European projects and exploitation of research results.
- (B) transfer of intellectual property to industrial operators.
- (C) sharing of developed code in applications through Free and Open-Source Software (FOSS) principles.

Regarding point (A), the three EU projects for which I was the Principal Investigator for CNR have received awards for achievements in technology transfer. Specifically, in ShareWork, the European Commission's Innovation Radar selected three implementation results developed by CNR (<https://innovation-radar.ec.europa.eu/resultbykeyword/sharework>). This Award recognizes the impact of CNR's scientific results within the project. As for the Robofoot project [PC14] was ranked among the top five projects of the Seventh Framework Programme (FP7-EU) for impact and implementation quality. To maximize the exploitation of results, my goal in recent years has been to seek funding for high-TRL actions, such as Large-Scale Demonstrators or actions related to translational research activities (e.g., leveraging Horizon Joint Undertaking calls). This choice is motivated by both cultural and managerial reasons. The Italian industrial fabric requires research entities to serve as benchmarks for the growth of the manufacturing sector. Guiding these companies through transformations of production systems by demonstrating high-impact project results creates added value for the country. From a managerial perspective, the high preparation of my group's human resources and the availability of robotic solutions accumulated over a decade of European projects allows us to realize industrial demonstrators relatively quickly, reducing startup times and energy for new initiatives. This, in turn, provides a strong financial lever for investing in PhDs and basic research.

I support knowledge sharing through Free and Open-Source Software (FOSS) paradigms (C). This approach allows the creation of large, collaborative working communities. My research group has activated 70 public Open-Source projects, while the CARI lab (<https://github.com/JRL-CARI-CNR-UNIBS>) has 84 active projects with an average of 50 users. The FOSS model offers several advantages, such as being more transparent, cost-effective, and robust and enabling rapid improvements through community collaboration. In this field, the open-source community formed around ROS (Robotic Operative System) has been a catalyst for transforming research worldwide.

B.1.3 Industrial and Institutional Roles

Regarding products and scientific qualifications, the most relevant aspects are consulting technical-scientific collaboration and support for companies. Specifically, I initiated industrial contracts worth approximately €600k. A long collaboration track with COMAU has been set in the last ten years.

I have participated in 12 evaluation commissions as an expert. Regarding my participation in evaluation committees, I have served on the monitoring committee of Lombardy Region overseeing the EcoCirc project; I am a CNR expert for the evaluation

of MISE projects, and I have been called upon as a member to evaluate EU-funded projects as well as international projects. I have served as the final evaluator for European initiatives.

At the national level, I have also participated in organizing some courses for SIRI (Italian Society of Industrial Robotics, www.siri.it). I was invited to speak at the MECPSE fair, a national reference for automation. Additionally, for MECPSE, I was a member of the committee that awarded the Innovation Awards in 2021. At the international level, I was invited to participate in several workshops organized by the European Robotic Forum, particularly in robotics-related events for advanced manufacturing. I was also invited to speak at the first B2B European robotics event.

In 2005, I founded PoliBrixia (www.polibrixia.it), a spin-off of the University of Brescia. This company is still active today and is a dynamic player in Research and Innovation in Industry 4.0 and Rehabilitation Robotics. I sold my shares when I became a permanent researcher at CNR-ITIA in 2011.

B.2 Host Institution

COMAU (CONsorzio MACchine Utensili) is an Italian multinational company specializing in industrial automation and robotics. Founded in 1973, COMAU is headquartered in Turin, Italy. The company offers automation solutions across various industries, such as automotive, aerospace, renewable energy, heavy industry, and logistics.

Key Areas of Expertise:

- **Robotics:** COMAU is a global leader in designing and producing industrial robots for welding, assembly, material handling, and painting. Their robots are used in high-precision applications and can be customized for different industries.
- **Industrial Automation:** The company provides automation solutions, including assembly lines, automated manufacturing systems, and equipment for body-in-white (the stage of car manufacturing where the car body is assembled).
- **Digital Manufacturing:** COMAU integrates digital technologies such as IoT, AI, and data analytics into manufacturing processes, supporting Industry 4.0 initiatives to optimize production, maintenance, and resource management.
- **Sustainability:** COMAU creates solutions contributing to sustainability, including automation systems for renewable energy projects, particularly in wind energy, solar energy, and energy storage.

COMAU operates in numerous countries worldwide, including major markets in Europe, North and South America, and Asia. It has over 4,000 employees and a network of global innovation centers, offices, and production plants.

COMAU is known for its cutting-edge technology, innovation in robotics, and contributions to modernizing manufacturing and production processes across industries.

The track of innovation of COMAU is also demonstrated by the large number of EU projects in which it has been involved. As shown at <https://dashboard.tech.ec.europa.eu/>, COMAU has been involved in 46 EU projects from FP7, ranking the company at 111 positions over 5583 funded institutions in Europe from FP7. 34% of the project is ongoing, making COMAU one of Europe's most active Robotics companies in research.

COMAU gathered about 17M€ from FP7 from participating in such projects, which has allowed COMAU to be among the most advanced worldwide robotic innovators.

COMAU has a well-consolidated knowledge of the industrial robotics field and will contribute to the project with its products, laboratories, and research facilities, where advanced solutions are developed and extensively tested. Moreover, COMAU will provide support and competence in integrating new products and processes by adopting the latest hardware and software technologies.

B.3 Affiliation Institution and Research Organization

The National Research Council of Italy is the largest national public research organization, nation-wide distributed in a network of 80+ Institutes, cross-disciplinary fostering science and innovation in biotechnology, medicine, materials, environment and land, information and communications, advanced systems of production, judicial and socio-economic sciences, classical studies and arts.

CNR 8000+ people are committed to promoting, transferring, and improving research activities in the main sectors of knowledge and applications for the scientific, technological, economic, and social development of the Country.

Two different Institutes of CNR will follow the project:

- **STIIMA** - Institute of Intelligent Industrial Systems and Technologies for Advanced Manufacturing. Specifically, STIIMA conducts scientific research, development, technology transfer, training, and strategic roadmapping activities to contribute to enterprises' innovation, competitiveness, and sustainability and promote people's central role in enterprises and society. In particular, the activities aim to design intelligent systems, enabling industrial technologies, products, and processes that co-evolve dynamically over time to meet different social and market needs and support new production paradigms. Alongside the Institute's manufacturing expertise, it is clear that areas of primary interest are represented by the study of natural and synthetic fibers and their characterization, functionalization, and transformation processes, the development of solutions for ambient intelligence and ambient assisted living, the central role of the individual in the society of the future,

with emphasis on the issue of assistance and the social and labor inclusion of vulnerable groups, advanced applications in the field of agricultural robotics and non-destructive quality control in the agri-food sector, bioinformatics and ecological informatics and the circular economy issue. CNR-STIIMA is the reference CNR institute for production systems (factory automation and robotics, mechatronics, industrial and service robotics, microsystems, controls, diagnosis, monitoring and new generation supervising systems, virtual design methods of products, virtual and augmented reality, process and factory design, management and business paradigms). CNR-STIIMA is a major partner of the European Technological Platform ManuFuture initiative (STIIMA's Director is a member of the High-Level Group and of the Implementation Support Group), finalized in developing added high-value manufacturing based on research and innovation. CNR-STIIMA is a founding member of the EFFRA association (private party of Factories of the Future PPP), the Director being a member of the Industrial Research Advisory Group and the Ad-hoc Industrial Advisory Group. CNR-STIIMA is a founding member of euRobotics AISBL (a private party of SPARC).

- **ISTC** Institute for Cognitive Science and Technology. The Institute participates in the project through the Planning and Scheduling Technology Lab (PSTLab - <http://istc.cnr.it/group/pst>), a group founded in 1997 and focused on artificial intelligence techniques for automated and interactive problem solving, integrated planning and scheduling algorithms, effective scheduling heuristics, dynamic scheduling, constraint programming, user interactive mixed-initiative systems. The group has been involved for over ten years in direct collaboration with the European Space Agency (ESA) to deliver several mission operational software systems (MEXAR, RAXEM, MrSPOCK) and two software infrastructures that are currently Agency assets (APSI, a planning software platform, and GOAC, a robot control software for autonomy). Between 2002 and 2006, the group coordinating entity of the RoboCare project, an initiative of the Italian Ministry of Research and University, was one of the first projects in Europe on robotic companions for older adult assistance. Since the start of FP7, the PSTLab has broadened its research interests toward new application areas, acquiring competencies in robotics (particularly human-robot interaction), user evaluation (for example, older adult users), ambient assisted living (smart environments integrating sensors, middleware, and intelligent services) and cognitive systems for training. The group has also worked in the manufacturing domain on national and international projects.

C Innovativeness and originality

C.1 Context

Cognitive robotics has the potential to revolutionize manufacturing by enabling robots to not only execute tasks but also understand and adapt to complex, dynamic environments.

With cognitive capabilities, robots could autonomously make decisions, collaborate more effectively with human workers, and handle unexpected situations, significantly boosting manufacturing productivity, efficiency, and flexibility. For example, a cognitive robot could adjust its behavior in real time based on new data, optimize processes on the fly, or even learn new tasks without extensive reprogramming. **However, several hardware and software tools are still missing to realize this vision fully.**

On the hardware side, more powerful yet efficient processors are needed to handle the computational demands of real-time decision-making and learning, along with advanced sensors for better environmental perception and interaction. On the software side, there is a need for more sophisticated frameworks that integrate machine learning, natural language understanding, and advanced planning algorithms. Furthermore, developing scalable and flexible software architectures that allow easy reconfiguration and learning across diverse tasks remains a significant challenge in the field.

The disruptive innovation for **Plan4AIR** consists of the **systematic approach to cognitive robotics** to realize the first cognitive industrial robot able to interact with humans and solve every task autonomously based on chatting with the process expert.

C.2 Artificial Robotic Intelligence: the innovation and the rocket science

According to the applied nature of the funding schema, **Plan4ARI** aims first to deploy a cognitive robot that exploits the best from the fundamental research. Generative AI, deliberative AI, deep learning semantic sensing, machine learning for motion planning, and neural control are recent scientific breakthroughs in each research area.

No research has tried to combine the best abilities to allow the robot to achieve a step-change in its cognitive ability. Robots are not only AI flesh out but also complex systems; the balance between the different modules is the real key to success.

To do so, **Plan4ARI** aims at a **pragmatic** approach:

- We do not reinvent the wheel. We want to exploit the best solutions available in the field – and there is a huge quantity of high-quality tools in academia (open-source community) and off-the-shelves tools
- We exploit the MI.RA Dexter COMAU platform, which allows low-frequency async communication with the robot to move it and access the industrial automation connected to the robot. This will make the development of the modules for natural language programming and adaptive motion planning straightforward.
 - **NOTE: all the scientific outcomes will be independent of the COMAU robot controller and extendable to all industrial manipulators. The adoption from COMAU of the Plan4ARI platform will grant an essential advance**

to the Italian sector worldwide, allowing the Italian industrial robotic industry to be a breakthrough in the market

- We will perform **fundamental research** only where there is a need. We will innovate in the field of
 - generative AI for the Knowledge Base foundation model and the PDDL file.
 - sensor fusion of radar sensing with 3D camera systems (a completely new research field).
- We will perform **industrial research** to deploy the best and most competitive tools for cognitive robotics. We will be disruptive in the development of robust and high-quality methodologies in the field of:
 - deliberative AI tools integration with STNU
 - adoption of MCTS to optimize the task plan using virtual execution of the tasks.
 - sample-based informed set motion planner to allow runtime motion planning
 - fusion of sample-based techniques with high-performing MPC to grant robustness in the task execution and exploitation of residual degrees of freedom of the setup
 - Deploy the first industrial controller, exploiting neural control design and prescribed performance constraint control to grant robustness and safety in all working conditions.

It is worth noting that adopting AI techniques in robotics is now at a critical stage. The innovation of ideas claimed in this project could be not so disruptive in 6 months, i.e., at the project start. The generative AI tools are evolving with impressive velocity, and an update of the scientific plan will be necessary once the project starts.

C.3 Artificial Robotic Intelligence: a framework to cognitive robotics

One crucial innovation presented by **Plan4AIR** demonstrates that the insertion of the proper AI tools at the correct robot control level is the key to having a completely autonomous mobile manipulator able to solve the problem when asked to.

The project's goal is that the motion control framework is as important as the single methodology. The full-working system will solve complex applications in a considerably short time.

The project will demonstrate that mobile industrial robots can be real companions of human beings. The trustworthiness investigation will be crucial at this point. These investigations are always limited to laboratory setups and small payload robots. Only a few works deal with high-payload industrial robots²⁶, but with a very narrow application. The role of COMAU in this evaluation will allow us to achieve an extraordinary impact in such a research field. Accessing the experience of one of the largest and most innovative industrial robot producers worldwide and connecting with its vast network of users will substantiate the research dramatically.

- The psychological approach to the evaluation of the interaction will drive the deployment.

The work package WP6 is designed to evaluate the interaction and extract the critical aspects of the cognitive processes that human beings activate when collaborating with the robot.

This cognitive evaluation of the interaction is the counterpart of having cognitive robots, and it is a powerful tool for designing the task and motion planner.

The definition of the paths, the sequence of the task, and the modality followed by the robot when it must slow down to preserve safety or hand something to the operators are fundamental factors for the definition of awareness in human operators.

The control framework will integrate the outcome of the trustworthiness investigation.

The investigation will allow the deployment of task and motion planning rules to improve users' acceptance of such technologies.

²⁶M. Rehm, I. Pontikis and K. Hald, "Automatic Trust Estimation From Movement Data in Industrial Human-Robot Collaboration Based on Deep Learning," 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, 2024, pp. 11245-11251, doi: 10.1109/ICRA57147.2024.10610822.

D Plan

D.1 Budget plan

Subjects	Fundamental research	Industrial Research	Experimental development	Total
Host Institution costs	€ 153,750.00	€ 2,492,750.00	€ 637,625.00	€ 3,284,125.00
<i>direct cost</i>	€ 7,500.00	€ 100,000.00	€ 25,000.00	€ 132,500.00
<i>Structured personnel</i>	€ 115,500.00	€ 1,894,200.00	€ 485,100.00	€ 2,494,800.00
<i>Personnel to be Acquired</i>	€ -	€ -	€ -	€ -
<i>indirect costs</i>	€ 30,750.00	€ 498,550.00	€ 127,525.00	€ 656,825.00
Affiliation Institution costs	€ 96,365.00	€ 192,730.00	€ 96,365.00	€ 385,460.00
<i>direct cost</i>	€ -	€ -	€ -	€ -
<i>Structured personnel</i>	€ 77,092.00	€ 154,184.00	€ 77,092.00	€ 308,368.00
<i>Personnel to be Acquired</i>	€ -	€ -	€ -	€ -
<i>indirect costs</i>	€ 19,273.00	€ 38,546.00	€ 19,273.00	€ 77,092.00
Research Organization costs	€ 242,500.00	€ 510,000.00	€ 47,500.00	€ 800,000.00
<i>direct cost</i>	€ 4,000.00	€ 8,000.00	€ 3,000.00	€ 15,000.00
<i>Structured personnel</i>	€ 95,000.00	€ 190,000.00	€ 15,000.00	€ 300,000.00
<i>Personnel to be Acquired</i>	€ 95,000.00	€ 210,000.00	€ 20,000.00	€ 325,000.00
<i>indirect costs</i>	€ 48,500.00	€ 102,000.00	€ 9,500.00	€ 160,000.00
Total	€ 492,615.00	€ 3,195,480.00	€ 781,490.00	€ 4,469,585.00
<i>Structured personnel</i>	€ 287,592.00	€ 2,238,384.00	€ 577,192.00	€ 3,103,168.00

D.2 Co-financing aspects

See FISA_2024_00099_Allegato B

D.3 Time Schedule

D.3.1 GANTT

Note:

- Each **Activity** will be documented following the continuous reporting approach. The document will report on the advancement and the results of the project itself.
- If the activity generates a dataset, the consortium will evaluate if the dataset can be made publicly available.

The GANTT of the activities is reported in the Figure below.

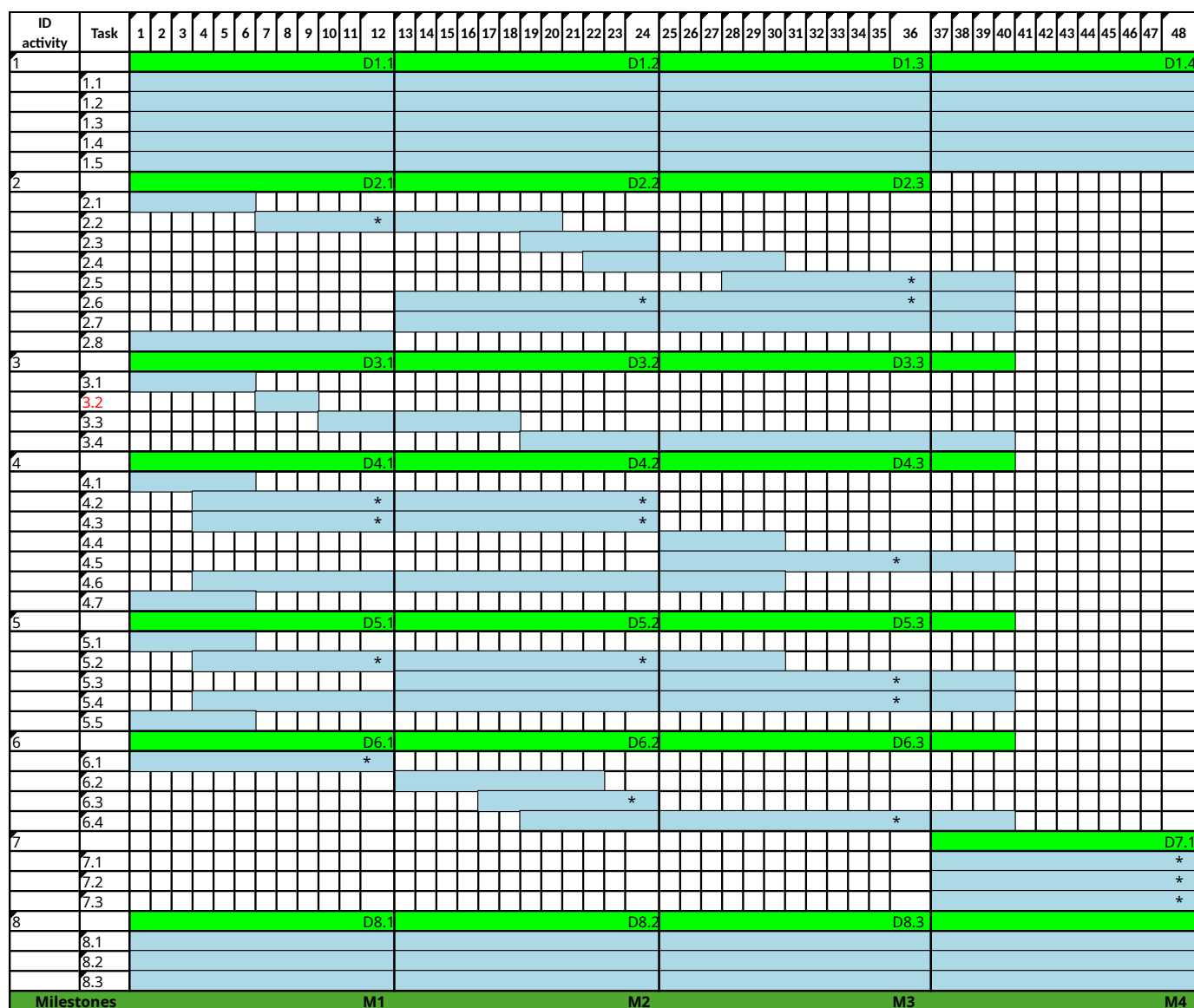


Figure 4 GANTT of the activities

D.3.2 List of Activities

ID activity	Description	Start month	Duration
1	Scientific, Technical, and Project Management	1	48

ID activity	Description	Start month	Duration
Objectives: The project Management will ensure that Plan4ARI contractual aspects are adhered to, including technical, financial, administrative, and consortium coordination and risk management, and will ensure that the project successfully achieves its specified objectives on time and within budget. Coordination issues, including contractual, legal, and administrative ones, are managed within this work package. Furthermore, ensuring dependable contact with the EU is an essential component of this WP over the project's duration. Task 1.1 General Management [CO FOUNDED ACTIVITY] M1-M48 This task foresees setting up and maintaining project quality, management procedures, reporting procedures, and templates for the consortium. Administrative project management: legal representation of the project, coordination, and development of implementation plans. Follow-up and reporting of project progress. Project Meetings coordination. Agreements preparation/maintenance. Setting up communication guidelines, rules, forms, and templates and maintaining internal communication levels and quality towards the financing institution. To oversee the promotion of gender equality in the project activities. Task 1.2 Financial management [CO FOUNDED ACTIVITY] M1-M48 This task foresees setting up financial guidelines, reporting procedures, and templates for consortium members, processing financial statements, preparing audit certificates, and preparing detailed financial plans. Task 1.3 Technical and Risk Coordination [CO-FOUNDED ACTIVITY] M1-M48 This task foresees technical activities management and progress follow-up. To ensure proper interaction between WPs. Intermediate project results (deliverables) and project outputs quality monitoring. To prepare and update the Data Management Plan. Manage technical risks and decision-making during project execution—initial risks monitoring and identifying new risks. Apply a contingency plan. Define and apply corrective measures. All WP leaders will support CNR as members of the Technical Committee. Task 1.4 Project Research Data Management [CO FOUNDED ACTIVITY] M1-M48 Data will be collected as part of the project implementation only after written informed consent of the enrolled subjects. Data will be fully anonymized, with subjects identified only by numbers. Study participants will be permitted to request removal of the coded data from the study without penalty. The research will be conducted on ethical implications according to the principles of widely recognized international texts and codes of practice (e.g., the Helsinki Declaration). Project results will fulfill the European GDPR and Data Protection Regulation superseding the Data Protection Directive. Task 1.5 Creation of Advisory Board [CO FOUNDED ACTIVITY] M1-M48 The Advisory Board will be set up in the first 6 Months of the project. The board will be continuously updated on the project's progress and provide a technical audit of the deployed solutions to balance the efficacy of the fundamental research with the exploitability of the delivered solutions. The board will comprise Academic experts external to the consortium and COMAU R&D top-figures.			
2	Low-code programming exploiting generative and deliberative AI	1	36

ID activity	Description	Start month	Duration
	Objectives: Plan 4ARI will use AI to decrease programming time and allow robot programming to process experts instead of robotic experts. AI will enable easy interaction between the user and the machine, pushing forward the spread of robotic technologies even to new sectors. The modules developed will be integrated into the COMAU MI.RA/Dexter framework. Task 2.1 Update of the scientific plan with the last state-of-the-art methodologies and tools [Fundamental Research] M7-M12 The generative AI tools and methods are impressively growing fast. A mandatory update of the scientific plan will be performed at the start of the project. The update will focus on: ■ The state of the art of the available tools, their costs, and performance ■ The state of the art of academic methodologies. The advancement could be overcome by the proposed approach in the next year. The technical approach will also be updated, maintaining the same objectives. Task 2.2 PDDL2 Generation for Robotic Tasks with Lightweight LLM from Natural Language [Fundamental Research] M13-M24 In recent years, robots have become increasingly prevalent in various fields, such as logistics, manufacturing, and healthcare, requiring adaptable and flexible task planning to handle dynamic environments and interactions. However, PDDL2 programming could be tedious and error-prone due to its complexity. Large language models (LLMs) have shown potential in aiding robot planning but are limited by resource demands. Task 2.3 PDDL2 to Behavior Tree Generation with time-variance modeling to integrate variability [Industrial Research] M18-M24 PDDL2 is the cornerstone for deliberative AI planning. However, the literature has not yet established temporal planners with time variance. Heuristic solutions are available but have not achieved robustness and widespread application. Simple temporal networks with uncertainty (STNU) and conditional simple temporal networks are the best candidates for allowing the transformation of the PDDL2 representation to behavior trees (BTs). Behavior trees have gained significant traction in robotics, offering a structured and scalable approach to managing complex robot behaviors. Originating from the video game industry, BTs have found relevance in robotics due to their ability to handle high-level decision-making and control in dynamic environments. A key player in this transition is the BehaviorTree.CPP library. This library has become the de facto standard in the Robot Operating System 2 (ROS2) ecosystem, mainly due to its inclusion in Navigation2. This integration has facilitated the development of sophisticated navigation behaviors, contributing to the broader adoption and standardization of BTs in robotics software design. It is worth noting that the translation BT does not solve the task planning problem. Still, it adjusts the planning domain to support adaptability in time-variable scenarios, which will be further addressed in the next task. Task 2.4 Task and Motion Planning Optimization in Digital Twin Environment [Industrial Research] M22-M30 The output of T2.2 represents the Task Planning problem using the Behavior Trees (BTs). The BTs are straightforwardly usable with Digital Representation of the context, for example, exploiting the ROS2 stack to integrate the motion planning virtualization and the human behavior estimation. In such a case, using Monte Carlo Tree Search methodologies, it is possible to virtually execute many epochs, simulate the task, and identify the optimal policy, given the function cost. Task 2.5 Optimal BehaviorTree Execution and Online Replanning [Industrial Research] M28-M36 The dispatcher for the actual execution will be interfaced with MI.RA/Dexter modules from COMAU. The system will have a module available to track the execution and the activation of the replanner as needed. The replanning method will be composed of two modules: ■ An Anytime planning module will continuously work concurrently with the execution to have an updated plan according to the scene ■ In the case of sudden and dramatic modification of the scene, a heuristic planner will be implemented for a fast re-start of the movement. In such a case, the plan will not be optimized, but it will be feasible. Task 2.6 Integration of the Task Planning Framework inside MI.RA/Dexter low-code [Industrial Research] M28-M36 Integrating the task planning pipeline inside the COMAU low-code programming tools will allow a straightforward digital twin of the setup. MI.RA/Dexter allows the creation of “dockerized” actions. This means that it is possible to integrate external code by creating a container and wrapping it through proper interfaces. This modular approach allows the integration of code that runs in a network of computation units. The deployment of the task planning module will follow a two steps approach: ■ Analysis of the SW modules, dependencies, and security constraints ■ Deployment of the proper actions to embed the code in the platform. The code embedding will be done eventually, reusing the code already deployed by the research organization. When the industrial requirements of such code are not met, the code will be re-engineered to grant maximum safety to the full setup. Task 2.7 Continuous Integration [Industrial Research] M13-M36 The SW development will follow a continuous integration paradigm. A cloud will be available to share the code among all the teams. Proper mechanisms for the versioning and remote checking will be implemented. The SW module will provide functional tests, and the SW development will be tracked according to the gold rules for SW deployment. Task 2.8 Benchmarking and Evaluation [Industrial Research] M7-M12 An extensive set of application examples will be defined in the first six months of the activity. The researchers will have the application goals as north-star to direct the deployment of the SW. The partners will agree upon the design of the experiments. The performance evaluation will be done by the Host Institution, which will invite operators and end-users to test the platform.		
3	Sensing for Context Awareness	1	24

ID activity	Description	Start month	Duration
Objectives: Plan4ARI will investigate using radar and 3D vision sensors to monitor the robot's working area, investigating the sensor fusion of multiple sources to provide robust operator tracking. The activity will be directly connected with the results of activity 4 and 5. The modules developed in Activity 3 will be integrated into MI.RA/Dexter COMAU framework. Task 3.1 Update of the scientific plan with the last state-of-the-art methodologies and tools [Fundamental Research] M1-M6 AI tools and methods are rapidly growing in semantic segmentation and pose estimation. A mandatory update of the scientific plan will be performed at the start of the project, focusing on: ■ The state of the art of the available tools, their costs, and performance ■ The state of the art of academic methodologies. If notable improvements emerge that may surpass the planned approach, adjustments to the technical approach will be made to maintain project objectives. Task 3.2 Radar sensor integration for scene monitoring [Industrial Research] M7-M9 In the market, radar technologies have started to offer the option of having 3D point clouds of objects inside the scene. The task will perform a detailed market investigation, and the more promising sensors will be acquired and tested. The sensor acquisition will then be packed inside a set of libraries that can be easily interfaced with the ROS2 network of devices. Task 3.3 Fusion of radar sensor data and camera data [Industrial Research] M10-M16 The challenge consists of fusing data from different sensors that are updated at different frequencies. This issue has been investigated mainly in automotive to ensure safe driving. The most common approach uses robust multi-object detection and tracking methods for moving objects. First, the radar and camera perform target detection independently, and the detection results are correlated in the image plane to generate a random finite set with an object type. Then, based on the Gaussian mixture probability hypothesis density (GM-PHD) algorithm framework, the tracking process will be refined through elliptic discriminant thresholds, an attenuation function, and simplified pruning methods. The module will be integrated as a functional block inside the MI.RA/Dexter Platform. Task 3.4 Semantic Segmentation [Industrial Research] M17-M24 The state-of-the-art semantic segmentation of images for identifying the Cartesian pose of objects involves several advanced techniques, mainly combining deep learning for segmentation and pose estimation approaches. MI.RA/Dexter already provides a few modules for pose estimation, and the host institution will put significant effort into bringing such modules to dramatically better performance. Deep Learning for Semantic Segmentation is often used for the segmentation, and modern systems utilize mainly Convolutional Neural Networks (CNNs) and Transformers to perform pixel-wise classification of images. Some key methods and architectures include Fully Convolutional Networks (FCNs), U-Net, DeepLab (v3+), and Mask R-CNN. Concerning 3D Object Pose Estimation, the task is generally tackled using Keypoint Detection Methods (like PoseCNN), or 6D Pose Estimation (like PVNet and PoseRBPF). The recent trends and techniques are mainly focused on the adoption of Transformers for Segmentation and End-to-End Pose Estimation Models. Methods such as "PoseNet" and "Single-shot pose estimation networks (SSD-6D)" allow simultaneous object detection, segmentation, and pose prediction in one unified framework. Also, "Graph Neural Networks (GNNs)" are extremely promising. The task will set up a benchmarking environment for testing the most promising state-of-the-art methods. The dataset will be created by selecting the most common requests from the COMAU users. The highest-performing algorithm will be added to the MI.RA/Dexter platform.			
4	Local and Global Autonomous Motion Planning	1	36

ID activity	Description	Start month	Duration
Objectives: a key aspect of modern collaborative robotic application is advanced motion planning. Plan4ARI aims to develop and deploy motion planners able to plan trajectories for a team of manipulators, in a collaborative dynamic environment. The modules will be integrated in the MI.RA/Dexter COMAU framework. Task 4.1 Update of the Scientific Plan with the latest State-of-the-Art Methodologies and Tools for Motion Planning [Fundamental Research] M1-M6 ■ Motion planning in robotics has seen rapid advancements, especially in real-time adaptability and high-dimensional search efficiency. This task will analyze the latest algorithms and frameworks, including their performance, computational demands, and scalability. Task 4.2 Informed Sample-Based Optimal Motion Planning for teams of mobile manipulators in dynamic environments [Industrial Research] M4-M24 The task deals with two goals. ■ It aims to create a mixed sampling strategy combining global and local informed sampling for asymptotically optimal sampling-based path planners. Local informed sampling oversamples the neighborhood of the current solution to quickly reach a local optimum, while globally, informed sampling guarantees asymptotic optimality. ■ We propose an asymptotically optimal algorithm using the mixed sampling strategy and dynamically adjust the trade-off between global and local sampling. We expect this approach to outperform state-of-the-art planners, such as Informed-RRT*, on different class problems. The methodology will then be integrated into an anytime planning framework to quickly provide a suitable solution in complex, high-dimensional search spaces. The core idea consists of building a graph that connects paths to the same goal, allowing the exploitation of pre-calculated paths. When an obstacle blocks the current path, it looks for a new one by trying to connect the current path to the other available ones. If a connection is found, it can search for better solutions during the remaining time. Task 4.3 Model Predictive Control for Optimal Adaptive Micro-interpolation [Industrial Research] M4-M24 The robot micro-interpolator running on the Core IPC (Figure 1) will exploit an MPC methodology. The MPC has the property of being able to predict future states in a predefined time window. In other words, it is possible to continuously create a queue of references for the low-level control with the dimension of the time window. MPC uses a dynamic model of the robot to predict its future states (e.g., positions, velocities, and accelerations) over a finite time horizon. This allows it to generate smooth and feasible trajectories for the robot to follow. Indeed, the MPC can handle Constraints. Industrial robotics usually has constraints on joint limits, velocity, acceleration, and even workspace boundaries. MPC can naturally incorporate these constraints into the optimization process, ensuring that the generated trajectory respects these physical limits. The resulting trajectory will be optimal for the selected cost function (typically including terms for trajectory tracking error, energy consumption, and smoothness). This ensures that the robot follows the desired trajectory as accurately and efficiently as possible. Modern advancements in computational power and optimization algorithms make it feasible to implement MPC in real-time, even for high-speed robotic systems. MPC can continuously update the control actions based on real-time feedback, making it robust against disturbances and model inaccuracies. The MPC will allow an easy and effective interface with the sensor for visual servoing while offering both Cartesian and joint interfaces. The MPC will be designed by exploiting the tools offered by COMAU to virtualize the robot. Task 4.4 Integration of the Motion Planning Framework inside MI.RA/Dexter low-code [Industrial Research] M25-M30 Integrating the motion planning pipeline inside the COMAU low-code programming tools will allow a straightforward digital twin of the setup. MI.RA/Dexter allows the creation of “dockerized” actions. This means that it is possible to integrate external code by creating a container and wrapping it through proper interfaces. This modular approach allows code integration in a network of computation units. The deployment of the task planning module will follow a two steps approach: ■ Analysis of the SW modules, dependencies, and security constraints ■ Deployment of the proper actions to embed the code in the platform. The code embedding will be done eventually, reusing the code already deployed by the research organization. When the industrial requirements of such code are not met, the code will be re-engineered to grant maximum safety to the full setup. Task 4.5 Integration of the MPC trajectory micro-interpolator inside the COMAU Core IPC [Deployment] M31-M36 Integrating the MPC inside the COMAU Core IPC will be a technicality that will require strong teamwork between COMAU and the Research Organization teams. The need for the module to support both a low-frequency async communication channel towards the external world and a hard-real real-time communication channel towards the robot will introduce a complex SW architecture. In addition, the MPC should also offer an interface for synchronous input communication. The MPC will mainly be designed to safely change the speed override of the robot according to its working condition. Task 4.6 Continuous Integration [Industrial Research] M4-M36 The SW development will follow a continuous integration paradigm. A cloud will be available to share the code among all the teams. Proper mechanisms for the versioning and remote checking will be implemented. The SW will provide functional tests, and the SW development will be tracked according to the gold rules for SW deployment. Task 4.7 Benchmarking and Evaluation [Industrial Research] M1-M6 An extensive set of application examples will be defined in the first six months of the activity. The researchers will have the application goals as north-star to direct the deployment of the SW. The partners will agree upon the design of the experiments. The Host Institution will evaluate the performance by inviting operators and end-users to test the platform.			
5	Neural Robust Motion Control	1	36

ID activity	Description	Start month	Duration
Objectives: Neural control algorithms constitute a frontier for robotic control but with a limited number of applications. Plan4ARI aims to investigate and integrate the last neural control algorithms for robot low-level control going beyond the current industrial SOTA. The modules developed will be integrated into MI.RA/Dexter COMAU framework. Task 5.1 Update of the scientific plan with the last state-of-the-art methodologies and tools [Fundamental Research] M1-M6 AI tools and methods are rapidly growing in semantic segmentation and pose estimation. A mandatory update of the scientific plan will be performed at the start of the project. The update will focus on: ■ The state of the art of the available tools, their costs, and performance ■ The state of the art of academic methodologies. The advancement could be overcome by the proposed approach in the next year. In such a case, the technical approach will also be updated, maintaining the same objectives. Task 5.2 Design of the neural controller with prescribed performance constraint architecture [Industrial Research] M4-M30 As discussed in the dedicated section, the team will conceive and implement a neural controller with the prescribed performance constraint architecture. The implementation of such a control law will follow the standard design procedures: ■ The mathematical model considering the lumped-elastic effect will be investigated and deployed ■ The control law will be designed, and the stability will be proven ■ The test will be primarily done using ad-hoc simulators in MATLAB, exploiting the wide framework developed in COMAU in the last 30 years. Once the control is confirmed and delivered, its implementation in C++ will be performed. First, the implementation will move small mechanisms controlled by a ROS2 architecture. The implementation in the COMAU controller will be in the following task. Task 5.3 Integration of the Motion Planning Framework inside the Open IPC [Industrial Research] M12-M36 Integrating the motion controller inside the COMAU Open IPC will allow a straightforward digital twin to be set up. MI.RA/Dexter allows the creation of “dockerized” actions. This means that it is possible to integrate external code by creating a container and wrapping it through proper interfaces. This modular approach allows code integration in a network of computation units. The deployment of the task planning module will follow a two steps approach: ■ Analysis of the SW modules, dependencies, and security constraints ■ Deployment of the proper actions to embed the code in the platform. The code embedding will be done eventually, reusing the code already deployed by the research organization. When the industrial requirements of such code are not met, the code will be re-engineered to grant maximum safety to the full setup. Task 5.4 Continuous Integration [Industrial Research] M4-M36 The SW development will follow a continuous integration paradigm. A cloud will be available to share the code among all the teams. Proper mechanisms for the versioning and remote checking will be implemented. The SW will provide functional tests, and the SW development will be tracked according to the gold rules for SW deployment. Task 5.5 Benchmarking and Evaluation [Industrial Research] M1-M6 An extensive set of application examples will be defined in the first six months of the activity. The researchers will have the application goals as north-star to direct the deployment of the SW. The partners will agree upon the design of the experiments. The Host Institution will evaluate the performance by inviting operators and end-users to test the platform.			
6	Human-Robot Trustiness Evaluation and Assessment	7	36
Objectives: Plan4ARI develops a tool to support manufacturers in assessing the workers' engagement when working with an AI-powered collaborative robot. T6.1 Subjective Evaluation: Design of the Questionnaire [Industrial Research] M7-M9 The goal is to investigate trust in a workstation, feelings, perceptions of the relationship between robot and operator, and other information. The following points are of interest: i) Scholarship/skills (attended classes about the workstation, about cobot, safety included); ii) Trust in the safety of the workstation, i.e., trust of being not physically damaged by the motion of the cobot while collaborating with it; iii) trust in software for safety installed on cobot; iv) trust that workstation as cobot and operator effectively accomplishes the job; v) degree of relevance attributed to the fact that the machine is a cobot instead of “any other” machine; vi) degree of feeling cobot as just a tool or as something close to colleague concepts; vii) degree of feeling playing a role that is active/leading/equal to cobot or being driven by/subject to cobot; viii) mental/physical stress before, during and after shifts with cobot; ix) impressions after a shift. Such topics have been drafted considering the manufacturing scenario. Debating with operators will be included to define topics. Each topic is expected to lead to one or more questions/scale items. T6.2 Subjective Evaluation: Questionnaire Validation [Industrial Research] M10-M16 We will exploit their contact at the national and international levels to contact companies to reach out to at least 50 workers who will be asked to complete the evaluation form. An online tool will be prepared. CNR will coordinate the scientific part, analyze the data, proceed with the interview as needed, and manage the GDPR-related aspects. T6.3 Subjective Evaluation: Questionnaire Deployment [Industrial Research] M17-M24 According to the dataset acquired in Task 6.2, a statistical evaluation will be performed to identify the statistically robust items coherent with the target scenario. The final evaluation form will be deployed according to the number of people evaluated in Task 6.2. T6.4 Subjective Evaluation: Evaluation Campaign, dataset creation synthesis [Industrial Research] M24-M36 The final campaign will follow the approach in Task 6.2. The target consists of having +100 workers interviewed.			
7	Targeted Demonstrator	36	48

ID activity	Description	Start month	Duration
-------------	-------------	-------------	----------

Objectives: the goal of this activity is to demonstrate all the [Plan4ARI](#) technologies on a targeted demonstrator provided by COMAU. The projects will demonstrate technologies on three different demonstrators coming from different industrial sectors. All the three demonstrators will exploit the technologies developed in the project.

Task 7.1 Picking random objects from a warehouse [Deployment] M36-M48

The use case includes the manipulation of a huge variety of objects in a warehouse.

With the massive spread of the e-commerce the request for automatic applications in warehouses is increasing. The application requires the manipulation of objects of a completely different nature, such as soft objects, clothes, or without standardized packaging. Currently this activity is mainly manual, imposing with high production rate and repetitive tasks, causing high stress on operators. The working environment is extremely dynamics and doesn't allow the installation of fences or rigid layout. The application involves the use of industrial mobile manipulator to pick a certain object from position A and release the object in a position B.

The operator will teach the possible objects to be found to the robot.

The operator will teach the task to the robot with natural language, such as "hey robot! Pick the red t-shirt in box A and then put the t-shirt in box B".

The system will interpret the command and translate the instruction into the robot program.

A task planner will schedule the tasks to all the manipulators necessary to reach the goal.

The motion planner will plan the trajectories for all the manipulators involved in the task avoiding the known object in the scene. A network of sensors continuously monitors the scene tracking the presence of the operator. In case of possible collision, the motion planner will replan the trajectory to avoid any possible collision. The operators involved in the demonstration will be asked to answer the questionnaire developed in activity 6.

The challenges in such a scenario are:

- The easiness in the KB creation
- The robot accurate localization
- The accurate semantic segmentation and pose estimation
- The offline motion planning
- The online motion planning – if the human is present

Task 7.2 Remanufacturing of mechanical components [Deployment] M36-M48

Remanufacturing of components and devices is becoming a key aspect for making more sustainable the manufacturing sector.

Remanufacturing avoids waste production and reduces the request of raw materials. The use case will involve the remanufacturing of a mechanical device. The type of device will be decided during the project. The use case scenario involves a non-robotic expert which will guide the robot through natural commands such as "Hey robot! Remove cover A from the device using the tool C and B".

The system will automatically generate robot instructions involving the task and motion planner generating an optimal task sequence and optimal trajectories.

Each task could require different control targets:

- Force/Admittance setpoint in interaction tasks
- High performance in Cartesian Error Tracking
- ...

Thanks to the low-level neural controllers, and the advanced dynamics modelling, the robot interaction with the environment during the disassembly tasks will be handled with higher performance compared to the current standard for industrial robot controllers, i.e. low-level control developed in [Plan4ARI](#) will guarantee an extremely high precision in robot-environment interaction forces estimation, such as during parts separation, or during the machining of components.

The operator will be fully involved in the process of being free to enter and intervene in the robot working area.

The tracking system made by the network of sensors will continuously track the operator. In case the operator enters the robot working area the motion planner will replan the robot trajectories to avoid any possible collision. In case the trajectory replanning will not be possible, the task planner will dynamically reschedule the task sequence allocating the robot to a different task. This iterative interaction will continue until the remanufacturing task is ended. The operators involved in the demonstration will be asked to answer the questionnaires from activity 6.

The challenges in such a scenario are:

- The easiness in the KB creation
- The robot accurate localization
- The accurate semantic segmentation and pose estimation
- The sensor fusion to achieve safety and performance in localization
- The online motion planning
- The hard-real time motion planning (MPC) to avoid any issue
- The performance of the low-level control
- The users' acceptance level for the AI-powered robot

Task 7.3 Welding of plates in huge boat [Deployment] M36-M48

The production of cargo and cruise boats deeply requires the use of welding processes to join huge plates. The welding process is currently done manually directly onsite (i.e. on the boat). The working conditions are severe, exposing the workers to unsafe and unhealthy working environments. The high demand of manual labor is also pushing the naval production to countries with a lower cost of manpower. For these reasons automation and robotics can bring innovation to the sector. Many barriers are anyway present in this working environment, such the extremely complex, dynamic and cluttered environment which constitutes the production site of a huge boat. Obviously, given the huge dimensions of the boat, the welding robot will be a welding mobile manipulator. The use of fences will not be possible since welding mobile manipulator will move around the boat where it is required to do the task. As for the other use cases, even in this case all the modules developed in [Plan4ARI](#) will be extensively used. An operator, expert about the welding process,

ID activity	Description	Start month	Duration
8	Communication and Dissemination	1	48
Objectives: (i) To continuously communicate the project activities and adequately disseminate results linked to the project exploitation plans. (ii) To conduct a proper market analysis and consequent management of project exploitable results supported by business case methodologies. (iii) To manage the exploitation plan, considering business plans, decisions, and agreements backed by protected IPR and suitable business models.			
Task 8.1 Communication activities M1-M48 This task will provide channels to communicate project objectives, status, progress, and results to the public. Essential elements of project communication are the project website, which is to be launched in M1 and updated throughout the project with news, information, and public deliverables produced by the project. Public Communication Materials – consisting of press releases, a project presentation, a digital project poster, a brochure to be adapted/printed on demand, and project videos related to project concepts and use case demonstrators. News and posts on digital media – the publication of news and posts will be planned and reported regularly.			
Task 8.2 Dissemination activities M1-M48 This task concerns disseminating project objectives, scientific relevance, and exploitable applications; detailed results will be planned. Dissemination actions that will be considered are Workshops, Seminars, Scientific publications, and primarily in charge of academic research institutions. Industries and tech providers will conduct trade fairs and exhibitions. All partners will contribute to the definition of the dissemination activities.			
Task 8.3 Market analysis and Exploitation plan M1-M48 This task will establish and maintain an exploitation culture throughout the project. This will be done through systematic exploitation activities during the project, accelerating the market adoption of project results. Critical elements of exploitation activities are cost and market analysis, which assesses the market potential and impact of project results tailored to partner exploitation channels. This analysis includes business modeling and potential future capital requirements associated with scaling up. Identification of Exploitable results, a project exploitable results table to identify, synchronize, and conduct efficient planning and management of the project foreground. Actions will be identified to develop and execute exploitation activities associated with each result, and management of the collective project foreground will be a part of periodic reporting and project meetings. An Exploration plan includes a study of the exploitation potential of each project result and the exploitation strategy, as well as establishing an appropriate calendar. Patent surveys and filing ongoing patent surveillance studies will allow for correctly managing the project's patentable results. Potential patents will be managed following the roles and priorities of the Exploitation Agreement.			

D.3.3 List of Deliverables

- D1.1 Management Report, M12
- D1.2 Management Report, M24
- D1.3 Management Report, M36
- D1.4 Management Report, M48
- D2.1 Scientific Plan and First Running Mockup of the Low-Coding Module, M12
- D2.2 First Running Time-Variance Aware Planner Running in MI.RA/Dexter Environment, M24
- D2.3 Optimal Task and Motion Planning with online replanning module, M36
- D3.1 Scientific Plan and First Running Mockup for the sensing module, M12
- D3.2 Sensing Methodologies and First Prototypes Release, M24
- D3.3 Sensing Modules integrated in the COMAU IPC, M36
- D4.1 Scientific Plan and First Mockup of the Motion Planning Modules, M12
- D4.2 Motion Planners Modules Prototypes, M24
- D4.3 Motion Planners Modules Prototypes integrated in the COMAU IPC, M36
- D5.1 Scientific Plan and First Mockup of the Neural Control Architecture, M12
- D5.2 First Release of the Neural Controller Prototype, M24
- D5.2 Neural Control Architecture integrated in the COMAU IPC, M36
- D6.1 Scientific Plan and First Questionnaire Release, M12
- D6.2 Second Questionnaire Release, M24
- D6.3 Campaign Evaluation Report, M36
- D7.1 Demonstrators
- D8.1 Exploitation, Communication, Dissemination Report, M12
- D8.2 Exploitation, Communication, Dissemination Report, M24
- D8.3 Exploitation, Communication, Dissemination Report, M36

D.3.4 List of Milestones

- Milestone 1, M12 (Mean of verification: D1.1, D2.1, D3.1, D4.1, D5.1, D6.1, D7.1, D8.1)
- Milestone 2, M24 (Mean of verification: D1.2, D2.1, D3.2, D4.2, D5.2, D6.1, D7.2, D8.2)

- Milestone 3, M36 (Mean of verification: D1.3, D2.1, D3.3, D4.3, D5.3, D6.1, D7.3, D8.3)
- Milestone 4, M48 (Mean of verification: D1.4, D2.1, D3.4, D4.4, D5.4, D6.1, D7.4, D8.4)

The Scientific Coordinator

Nicola Flocchi